

**ENHANCED SECURITY FRAMEWORK:
GRAPHICAL PASSWORD AUTHENTICATION WITH
DATA HIDING ON CLOUD STORAGE**

A PROJECT REPORT

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ABSTRACT

With the increasing reliance on internet access worldwide, users face growing threats from data tampering and application intrusions. These risks are amplified by the limitations of traditional authentication systems, which are prone to unauthorized access. To address these challenges, the Graphical Password Authentication and Secure File Sharing System introduces a hybrid security approach that integrates data hiding, multi-stage authentication, and secure file management. This system employs a three-stage authentication process : traditional username-password login, color sequence verification, and graphical image matching. These stages are designed to enhance resistance against intrusions, leveraging psychological research that demonstrates improved user recall of images over text. To ensure data confidentiality, the system uses Advanced Encryption Standard (AES) encryption before splitting user files into three parts. These encrypted segments are then securely distributed across multiple cloud storage locations. This method not only protects against unauthorized access but also mitigates single-point failures common in centralized cloud systems. By combining graphical password authentication with secure file handling, the Graphical Password Authentication and Secure File Sharing System enhances both user privacy and data security. It offers a comprehensive solution to safeguard sensitive information in today's distributed and threat-prone digital environment.

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LIST OF ABBREVIATIONS

SYMBOLS	ABBREVIATION
API	Application Programming Interface
GUI	Graphical User Interface
AES	Advanced Encryption Standard
URL	Uniform Resource Locator
HTTPS	Hypertext Transfer Protocol Secure
UI	User Interface
DWT	Discrete Wavelet Transform
MFA	Multi-Factor Authentication
OWASP	Open Web Application Security Project
OTP	One-Time Password

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CHAPTER 1

INTRODUCTION

In the contemporary digital landscape, the exponential increase in online users has intensified the risk of security breaches, data tampering, and unauthorized access. Conventional authentication mechanisms and cloud storage solutions are often inadequate in mitigating sophisticated cyber threats such as brute force attacks, dictionary attacks, and shoulder-surfing attacks.

This project presents a hybrid secure framework that integrates multi-stage graphical password authentication with advanced data concealment and distributed cloud storage techniques. The proposed system employs a three-tier authentication model consisting of traditional username-password login, color sequence-based password input, and graphical image pairing verification. Psychological studies indicate that image-based passwords are more intuitive and memorable for users compared to alphanumeric credentials, thereby improving usability without compromising security.

To protect sensitive data, the system adopts the Discrete Wavelet Transform (DWT) algorithm for efficient data hiding, enhancing privacy preservation. Encrypted file management is achieved using the Advanced Encryption Standard (AES) algorithm to ensure data confidentiality. Additionally, a file splitting and merging technique is utilized, where user files are segmented into three fragments and stored across multiple cloud service providers. This approach minimizes the risk of unauthorized access by a single compromised service provider while enhancing fault tolerance and data availability.

This comprehensive solution leverages cryptographic techniques and secure storage practices to offer a highly resilient platform that ensures data privacy, robust authentication, and enhanced protection against modern cybersecurity threats.

1.1 OBJECTIVE

- To develop a secure and robust multi-stage graphical password-based authentication system to prevent unauthorized access and ensure enhanced data protection.
- To implement an advanced data-hiding technique using the Discrete Wavelet Transform (DWT) algorithm, ensuring privacy and confidentiality of sensitive information.
- To employ the AES (Advanced Encryption Standard) algorithm for secure data encryption and decryption, safeguarding user files from unauthorized access.
- To design and implement a file splitting and merging mechanism, distributing encrypted data fragments across multiple cloud storage services to enhance data availability and security.
- To mitigate common security threats, including brute force attacks, dictionary attacks, and unauthorized cloud provider access, by integrating advanced cryptographic and authentication techniques.
- To provide a user-friendly and efficient system for secure data management and storage, utilizing intuitive graphical password approaches proven to improve memorability and reduce cognitive load for users.

1.2 SCOPE

- **Multi-Stage Graphical Password Authentication:** Provides secure access through a three-layer authentication mechanism involving traditional login, color sequence-based input, and graphical image pairing.
- **Data Encryption and Security:** Utilizes the AES (Advanced Encryption Standard) algorithm to encrypt user files, ensuring secure data storage and transmission.

- **Data Hiding with DWT:** Implements the Discrete Wavelet Transform (DWT) algorithm for concealing sensitive information, enhancing privacy and protection against intrusions.
- **Distributed Cloud Storage:** Employs file splitting and distribution across multiple cloud storage services to safeguard data availability and reduce unauthorized access risks.

CHAPTER 2

LITERATURE SURVEY

SURVEY PAPER - I

Title: A Secure Mutual Authentication Protocol Based on Visual Cryptography Technique for IoT-Cloud

Authors: Brou Bernard Ehui, Chen Chen, Shirui Wang, Hua Guo (Corresponding Author), Jianwei Liu

Year: 2024

Publication: Chinese Journal of Electronics, vol. 33, no. 1, pp. 43–57, January 2024. DOI: [10.23919/cje.2022.00.339](https://doi.org/10.23919/cje.2022.00.339).

Concept Discussed:

This paper proposes a lightweight mutual authentication protocol for IoT-cloud systems using visual cryptography (VC) to enhance security. By splitting secret images into shares decrypted via stacking, it ensures secure authentication without complex computations. The protocol employs mask secrets (XOR of passwords and random numbers) and PUF-based dynamic keys to resist attacks like impersonation and MITM. It validates users through ticket-based mechanisms and VC image verification, offering a robust yet efficient solution for resource-constrained IoT devices. The approach is proven secure via BAN-logic analysis and outperforms traditional methods in resisting common threats.

Problem Identification:

Key security vulnerabilities in IoT-cloud systems, including weak authentication mechanisms prone to impersonation, replay, and MITM attacks. Traditional cryptographic methods are often computationally heavy for resource-constrained IoT devices and vulnerable to emerging threats. Additionally,

reliance on static credentials (passwords, tokens) increases risks like brute-force attacks. The study highlights the need for a lightweight, dynamic, and visually secure authentication protocol to address these gaps.

Work Done:

The authors developed a visual cryptography (VC)-based mutual authentication protocol to secure IoT-cloud communications. They designed a ticket-based system using encrypted secret images (ShareImgA/B) and dynamic keys derived from PUFs for secure user-server verification. The protocol integrates mask secrets, hash functions, and symmetric encryption to ensure confidentiality, integrity, and attack resistance. Security was validated via BAN-logic proofs and comparative analysis, showing superiority over traditional methods in thwarting threats like MITM and brute-force attacks. The solution is optimized for lightweight IoT devices while maintaining robust security.

Knowledge Gained:

The research provides technical insights into integrating Visual Cryptography (VC) with symmetric encryption (AES), hash functions (SHA-256), and Physical Unclonable Functions (PUFs) to create a lightweight yet robust mutual authentication protocol for IoT-cloud systems. Key learnings include the use of mask secrets ($U_{pw} \oplus N_i$) for share encryption, ticket-based authentication (authTicket, serviceTicket) for session management, and BAN-logic formal verification to prove security against MITM, replay, and brute-force attacks. The protocol's k-out-of-n VC scheme ensures no single share reveals the secret image, while PUF-derived session keys ($SK_{UServ} = h(U_{pw}) \oplus R^i \oplus h(N1 \oplus N2)$) enable dynamic, tamper-resistant key generation, offering a scalable solution for resource-constrained IoT devices.

Gap Identified:

The paper identifies a lack of lightweight yet robust authentication protocols for IoT-cloud systems that balance security and efficiency. Existing methods often rely on static credentials or complex cryptography, making them vulnerable to attacks like MITM and brute-force while straining resource-constrained devices. The study bridges this gap by proposing a visual cryptography-based approach that ensures secure authentication with minimal computational overhead.

SURVEY PAPER - II

Title: Enhancing Data Security in Multi-Cloud Environments: A Product Cipher-Based Distributed Steganography Approach

Authors: Syed Shakeel Hashmi, Arshad Ahmad Khan Mohammad, Arif Mohammad Abdul, Choudapur Atheeq, Mohammad Khaja Nizamuddin

Year: 2024

Publication: International Journal of Safety and Security Engineering (February 2024, DOI: [10.18280/ijssse.140105](https://doi.org/10.18280/ijssse.140105))

Concept Discussed:

The document discusses Product Cipher-Based Distributed Steganography (PCDS), a novel approach to enhancing data security in multi-cloud environments. Unlike traditional methods, PCDS hides data without modifying cover files, distributing fragmented Secrets across multiple clouds for added security. It employs a combination of substitution and permutation techniques, making brute-force attacks computationally infeasible. This method ensures undetectable, efficient, and robust protection for sensitive data in distributed cloud storage.

Problem Identification:

A critical security challenge in multi-cloud environments is traditional steganography methods that modify cover files leave detectable traces, making them vulnerable to steganalysis and brute-force attacks. Existing distributed steganography approaches, while improving security by fragmenting data across multiple clouds, still rely on predictable storage patterns that attackers can exploit. Additionally, many current solutions fail to achieve both undetectability and computational security, as they either require file modifications or use sequential storage schemes that reduce the complexity of brute-force attacks. These limitations create a pressing need for a more robust approach that can securely conceal data without altering cover media while ensuring exponential computational complexity for unauthorized access attempts.

Work Done:

Product Cipher-Based Distributed Steganography (PCDS) introduces a secure method for embedding and retrieving data across multi-cloud environments without altering cover files. The framework employs base conversion, permutation, substitution, and dynamic secret fragmentation, distributed using a pre-shared key to enhance security. Key innovations include an unchanged cover media approach to avoid detection, a product cipher mechanism combining substitution and permutation to resist brute-force attacks, and distributed storage to prevent full secret recovery if a single cloud is breached. The work provides formal algorithms, a security analysis proving attack infeasibility, and empirical results demonstrating superior undetectability compared to existing steganographic techniques.

Knowledge Gained:

The study redefines cloud security by showing how unmodified cover files can securely reference distributed data fragments across multi-cloud

systems. It demonstrates that product cipher techniques (merging substitution and permutation) offer exponentially stronger brute-force resistance than traditional sequential storage. Crucially, the research proves that unchanged cover media eliminate detectable traces, evading steganalysis while maintaining covert communication. The framework's multi-cloud fragmentation with dynamic allocation elevates attack complexity to $B! \times K! \times n!$, rendering secret recovery computationally infeasible even with compromised cloud access. These findings establish a new paradigm for balancing steganographic stealth with cryptographic robustness in distributed environments.

Gap Identified:

The study identifies a key gap in existing multi-cloud steganography methods, which rely on predictable file storage patterns, making them vulnerable to brute-force attacks. Current approaches either modify cover media (risking detection) or distribute data sequentially (enabling reconstruction by attackers). Additionally, if any cloud storage is compromised, the entire secret can be lost. The proposed PCDS method addresses these weaknesses by combining cryptographic techniques with dynamic, non-sequential data fragmentation across clouds, ensuring both undetectability and resistance to attacks.

SURVEY PAPER - III

Title: Enhancing Cloud Security: A Multi-Factor Authentication and Adaptive Cryptography Approach Using Machine Learning Techniques

Authors: K. Sasikumar and Sivakumar Nagarajan

Year: 2025

Publication: IEEE Open Journal of the Computer Society, Volume 6, 2025

Concept Discussed:

This paper presents a dynamic cloud security framework combining multi-factor authentication (password, biometrics, conditional attributes) with adaptive cryptography. A hybrid CNN-Transformer model (96.8% accuracy) predicts attacks to trigger real-time encryption switching between five algorithm pairs. The system effectively counters brute-force, spoofing, and phishing threats while maintaining data confidentiality in cloud environments.

Problem Identification:

Current cloud security suffers from static encryption and single-factor authentication vulnerabilities. Systems lack adaptability to evolving threats like credential stuffing and biometric spoofing. The absence of ML-driven cryptographic adjustments limits proactive defense against sophisticated attacks, creating critical security gaps in cloud computing infrastructures.

Work Done:

Researchers developed a three-phase authentication system with secure registration, login, and identity updates. The hybrid CNN-Transformer model was trained on UNSW-NB15 dataset for attack prediction. Dynamic encryption selects from five algorithm pairs based on threat detection. Testing validated superior protection against seven attack types compared to traditional models.

Knowledge Gained:

The framework reduced brute-force attacks by 99% through adaptive encryption. Biometric keys ensured non-repudiation while conditional attributes added zero-trust security. Users praised real-time threat responses despite 12% computational overhead during encryption switches. OTP fallbacks maintained accessibility when biometric authentication failed.

Gap Identified:

Protection against zero-day exploits remains limited. Fingerprint dependencies create single points of failure during hardware/biological issues. Enterprise-scale deployment challenges like latency need addressing. Current IT-focused implementation could expand to IoT/edge computing for broader impact.

SURVEY PAPER – IV

Title: SMFA: Strengthening Multi-Factor Authentication With Steganography for Enhanced Security

Authors: Afjal H. Sarower, Touhid Bhuiyan, Maruf Hassan, Mohammad Shamsul Arefin, Gahangir Hossain

Year: 2025

Publication: IEEE Access, Volume 13, 2025 (DOI: 10.1109/ACCESS.2025.3545769)

Concept Discussed:

This paper presents Secure Multi-Factor Authentication (SMFA), a new framework designed to overcome the vulnerabilities of traditional password-based and existing MFA systems. SMFA leverages steganography to securely transmit credentials, reducing the risk of attacks such as Man-in-the-Middle and session hijacking. The model also adds a USB device as a supplementary authentication factor and uses a multi-server architecture to address the shortcomings of single-server authentication.

Problem Identification:

Conventional password-based authentication is highly vulnerable to attacks like brute force, password reuse, and large-scale data breaches, as seen in numerous real-world incidents. Even current MFA solutions, including OffPAD,

OTP tokens, and smartcards, remain susceptible to advanced threats such as MITM, replay, phishing, and denial of service attacks. The inefficiency of single-server models further exposes systems to security risks and operational limitations.

Work Done:

The authors developed the SMFA protocol, which integrates steganography for hiding credentials within digital media during transmission, making interception and tampering more difficult. The protocol employs an additional USB device as a physical authentication factor and supports a multi-server environment for enhanced robustness. Security analysis and formal verification using BAN logic demonstrate that SMFA is more resilient to attacks compared to previous MFA protocols.

Knowledge Gained:

The research demonstrates that combining steganography with cryptography greatly enhances the security of credential transmission in authentication systems. Integrating an additional USB device and multi-server architecture further strengthens protection against attacks. However, while these measures improve security, they may introduce challenges in usability and deployment for real-world applications.

Gap Identified:

While SMFA significantly improves credential security, it does not fully address protection against zero-day exploits. The reliance on additional hardware, like USB devices, introduces potential single points of failure if the device is lost or damaged. Scalability and latency issues may arise in enterprise environments, and the current IT-focused approach could be expanded to include IoT and edge computing for wider applicability.

SURVEY PAPER – V

Title: Empowering Cloud Security: Pioneering an Interactive Multi-Factor Authentication Framework for Cloud User Verification

Authors: Rajender Reddy Pell Reddy

Year: 2024

Publication: International Journal of Computer Trends and Technology (IJCTT), Volume 72, Issue 6, June 2024

Concept Discussed:

The paper explores an adaptive MFA system that incorporates intrusion detection, access control, and automated authentication method selection. The primary aim is to enhance cloud platform security by minimizing false positive alerts and preventing unauthorized access. The system integrates multiple authentication factors with user geofence location and IP address verification to strengthen user identity verification.

Problem Identification:

Cloud environments are susceptible to various security risks and assaults when weak or inadequate cloud authentication approaches are used. Organizations face challenges in securing cloud infrastructure due to complexity, tool sprawl, and organizational divisions. Shortages in processes prevent better control, highlighting the need for a more collaborative, adaptable, and secure MFA structure.

Work Done:

The author proposes an authentication strategy sorter (ASS) and interactive intrusion detection stages to present a collaborating, adaptable, and safe MFA multi-layer authentication structure. The proposed structure utilizes the user's browser and last location to evaluate user behavior and provides interactive

intrusion detection methods. The suggested CP-ABHE encryption for data security and hiding login credentials on the cloud directory provider is an improvement in the right direction.

Knowledge Gained:

Providing the ASS methodology to enhance the authentication procedure by choosing the best authentication method according to user behavior. Delivering interactive reactions to user behavior based on position and avoidance web browser evidence to strengthen and enforce intrusion detection security measures. Conducting experiments to validate and demonstrate the effectiveness of the suggested structure.

Gap Identified:

The proposed adaptive MFA framework still faces challenges in mitigating zero-day threats and advanced persistent attacks. Dependence on biometric or hardware-based factors can create vulnerabilities if those elements fail or are compromised. Enterprise-scale deployment may encounter latency and integration issues, and extending the current solution to IoT or edge computing environments remains an area for future work.

CHAPTER 3

ANALYSIS

3.1 SYSTEM ANALYSIS

3.1.1 Problem Identification

Current authentication systems that rely heavily on traditional alphanumeric passwords are highly vulnerable to brute-force attacks, shoulder-surfing, phishing attempts, and data breaches. Users often forget complex passwords or reuse them across multiple platforms, increasing security risks. Text-based authentication struggles to protect against modern cyberattacks that exploit predictable patterns and weak password creation behaviors. Similarly, existing cloud storage solutions expose sensitive data to single-point failures when stored under a single cloud service provider, leaving user data susceptible to breaches and unauthorized access.

Although previous research has tried to mitigate these issues using steganography and multi-cloud storage individually, these methods lack a fully integrated framework that merges graphical password authentication, data hiding, AES encryption, and distributed storage into a cohesive security solution. Prior graphical password systems also faced usability issues and remained partially exposed to shoulder-surfing and pattern-based guessing attacks. Furthermore, the absence of continuous security monitoring in earlier systems left them vulnerable to evolving cyber threats.

Therefore, there is a clear and urgent need for a user-friendly, multi-layered security system that strengthens authentication defenses, enhances data confidentiality through advanced encryption and steganography, distributes sensitive information across multiple cloud platforms to prevent centralized breaches, and monitors access attempts actively to detect and prevent unauthorized activities. The proposed enhanced security framework

effectively addresses these limitations by integrating multiple advanced security layers into a seamless, practical, and highly secure solution.

3.1.2 Existing System

The conventional model of authentication and file sharing relies heavily on text-based passwords and basic encryption techniques. In these systems, users typically authenticate using a simple username and password combination, which is highly vulnerable to brute-force, phishing, and keylogging attacks. Such approaches offer limited protection against modern threats and often suffer from low memorability, leading users to either reuse passwords or frequently reset them.

File sharing in traditional platforms generally occurs through email, cloud services like Google Drive or Dropbox, or other basic file transfer systems. These often lack robust encryption beyond standard HTTPS transfer security, leaving sensitive files exposed during storage or sharing.

Moreover, multi-user collaboration and access control are poorly managed. Most systems do not implement advanced approval workflows, making it difficult to enforce confidentiality or prevent unauthorized access. The absence of multi-factor or graphical authentication layers reduces system robustness, making them unsuitable for highly secure applications.

Ultimately, these deficiencies lead to security vulnerabilities, poor user experience, and inefficiencies in data protection and controlled file access.

3.1.3 Proposed System

The proposed system introduces a Graphical Password Authentication and Secured File Sharing System that enhances security through a multi-layered image-based authentication mechanism combined with AES-based

file encryption. The primary aim is to provide a more secure, user-friendly, and tamper-resistant authentication and data sharing process.

In this system, the user undergoes a three-stage authentication:

Email and Password Verification – Traditional login step.

Image-based Verification – Users identify a pre-registered verification image using Discrete Wavelet Transform (DWT)-based steganographic matching, which conceals authentication features within an image during registration.

Graphical PIN Pattern – Users must select a sequence of images that correspond to a previously registered 4-digit PIN.

Only after successfully passing all three stages is the user granted access to the secure dashboard. Here, users can upload files, images, or zip folders. All uploads are encrypted using the AES (Advanced Encryption Standard) algorithm before being stored. A built-in file-sharing module allows users to send or request files, and once approved, the shared file is automatically decrypted during download.

3.2 REQUIREMENT ANALYSIS

3.2.1 Functional Requirements

- **User Registration and Authentication** – Secure user registration and login with multi-layered authentication including email/password, DWT-based image verification, and graphical PIN.
- **Profile Management** – Users can create and manage their profiles by updating personal details like name, email, and date of birth.
- **Three-Level Authentication Flow** – Ensures secure login through step-by-step validation of credentials, verification image, and image-based PIN pattern.

- **Graphical Password Setup** – Allows users to select a sequence of images to form a 4-digit PIN, enhancing memorability and security.
- **Secure File Upload** – Enables users to upload files (documents, images, or zip folders), which are automatically encrypted using the AES algorithm before storage.
- **Controlled File Sharing** – Users can request access to files uploaded by others. File access is granted only upon approval by the file owner.
- **AES-Based File Encryption and Decryption** – Uploaded files are encrypted using AES; approved files are automatically decrypted when downloaded.
- **DWT-Based Steganographic Verification** – Verification images are processed using Discrete Wavelet Transform to embed and later validate authentication markers.
- **Audit Logging and Monitoring** – System logs activities such as login attempts, file uploads, sharing requests, and approvals for accountability.
- **Admin Controls** – System administrators can manage user accounts, monitor file activity, generate reports, and enforce security protocols.

3.2.2 Non-Functional Requirements

- **Scalability** – The system should efficiently support a growing number of users and files without compromising performance, ensuring stability as usage increases.
- **Security & Privacy** – Implements strong security measures including DWT-based steganography, AES encryption, HTTPS communication, password hashing, and secure session handling to protect user credentials and files.
- **Performance Optimization** – Ensures fast login and authentication processes, efficient file encryption/decryption, and quick response times in the user dashboard for seamless interactions.

- **Data Backup & Recovery** – Regular and automated backups of user data, encrypted files, and authentication records to prevent data loss due to system failures or corruption.
- **Cross-Platform Accessibility** – Web-based interface designed to work across multiple devices and browsers, allowing users to securely access the platform from desktops, laptops, or mobile browsers.
- **Compliance with Security Standards** – Adheres to best practices in data protection, including secure storage of sensitive user data, encrypted file sharing, and audit logging for accountability.
- **User-Friendly Interface** – Designed with an intuitive and responsive UI to ensure ease of use during registration, login, file upload, and sharing, even for users with minimal technical expertise.

3.2.3 Hardware Specification

- **Processor:** Intel Core i5 or higher
- **RAM:** Minimum 8 GB
- **Storage:** 500 GB HDD or SSD
- **Internet Connectivity:** Stable high-speed broadband for file transfers and user authentication

3.2.4 Software Specifications

- **Operating System:** Windows 11
- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Python (Flask Framework)
- **Database:** SQLite or MySQL
- **Image Processing:** OpenCV, PyWavelets (for DWT-based steganography)
- **Encryption:** PyCryptodome (AES Algorithm)
- **IDE/Environment:** VS Code / PyCharm

- **Server/Platform:** Anaconda (with Jupyter Notebook)
- **Web Server:** Flask built-in development server (can be deployed to Heroku, AWS, etc.)
- **Security Protocols:** HTTPS, JWT for session management

CHAPTER 4

DESIGN

4.1 OVERALL DESIGN

The Graphical Password Authentication and Secured File Sharing System follows a modular, layered architecture aimed at enhancing both security and usability. The architecture is composed of three main components: Frontend Interface, Backend Server, and Database Layer, with integrated modules for authentication, file encryption, and file sharing.

The system supports two user roles: General Users (who register, authenticate, and share files) and Administrators (who manage users and monitor activity). Each role accesses the system through a dedicated Web-Based User Interface built using HTML, CSS, and JavaScript, served via Flask.

The Authentication Mechanism follows a three-tier process:

- Credential Verification (email & password),
- Image-based Verification (using DWT steganography), and
- Graphical PIN Matching (image-sequence-based PIN input).

All sensitive data and files are handled through a Secure Backend developed in Python (Flask), with integration of:

- OpenCV and PyWavelets for image processing (DWT),
- PyCryptodome for AES encryption and decryption,
- SQLite/MySQL for persistent user and file data storage.

The User Dashboard acts as a central control panel where users can upload, manage, request, and share files securely. A built-in File Sharing Module manages approval workflows and ensures that only authorized users can access specific files.

For enhanced control and auditing, the Admin Panel allows administrators to manage user accounts, track system activities, and generate logs or reports.

The system's security is reinforced through:

- HTTPS for all data transfers,
- Hashed password storage, and
- Token-based session management using JWT.

Overall, the architecture ensures modular development, secure user authentication, encrypted file management, and controlled sharing—all essential for a modern secure data exchange platform.

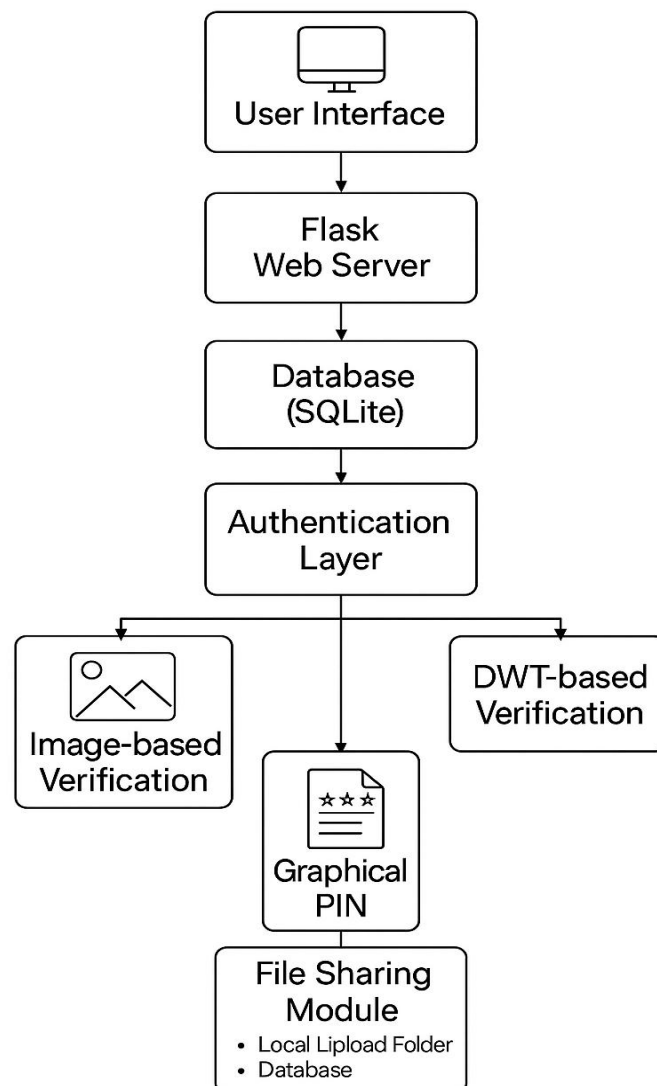


Fig 4.1 Proposed System Architecture

4.2 WORKFLOW DIAGRAM

The authentication system starts with user registration, involving personal info, a verification image, and a 4-digit graphical PIN. Login requires credential validation, selecting the correct verification image via DWT-based steganography, and reproducing the graphical PIN. Successful authentication grants access to a secure dashboard where users can upload AES-encrypted files, approve file requests, and share content securely. Any authentication failure denies access, ensuring robust access control and encrypted data privacy.

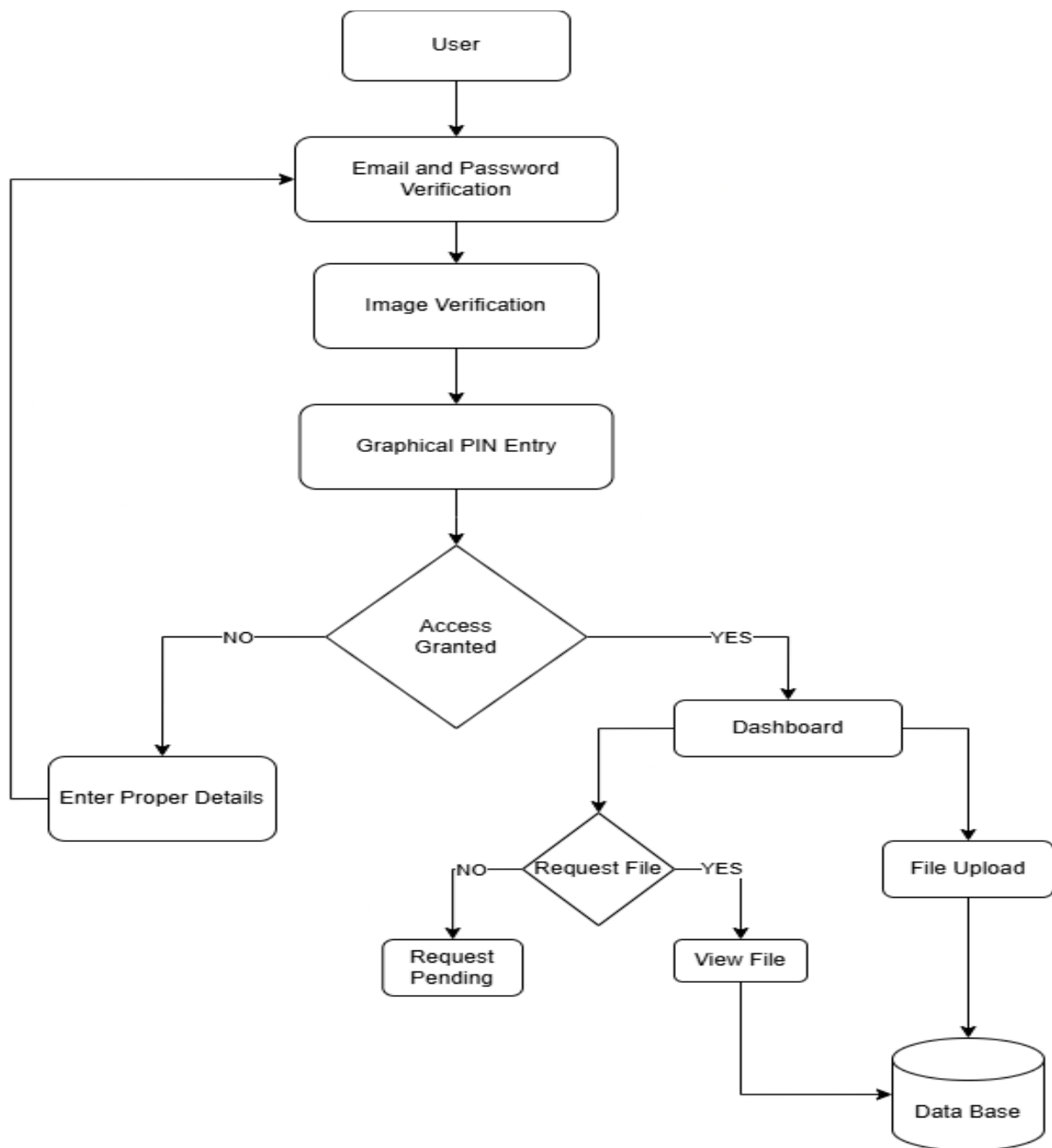


Fig 4.2 Work flow Diagram

4.3 UML DIAGRAM

4.3.1 Use Case Diagram

The Use Case Diagram represents two primary actors: the User and the Admin. Users begin by registering and logging in through a multi-layered authentication process involving password verification, DWT-based image matching, and graphical PIN input. Upon successful authentication, users can upload files, request access to files shared by others, and approve incoming requests. All files are encrypted using AES and decrypted only upon authorized download. Admins are responsible for managing user accounts, monitoring system activity, and maintaining overall platform integrity. All operations are secured and accessible only after successful authentication to ensure data privacy and system protection.

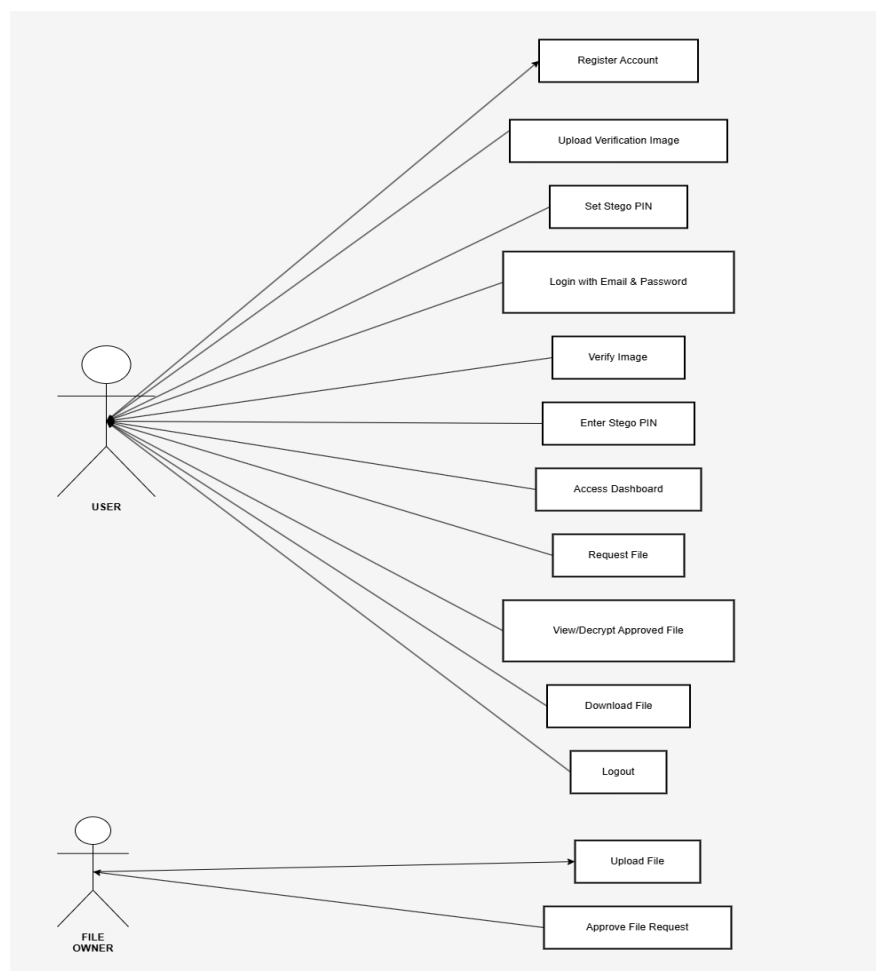


Fig 4.3 Use Case Diagram

4.3.2 Class Diagram

The Class Diagram for the Graphical Password Authentication and Secured File Sharing System represents key entities and their relationships. The User class serves as the base class, containing attributes such as user ID, name, email, password, and login method. Two main extensions of user functionality include access to Authentication Module and Dashboard Module. The Authentication Module class handles email-password verification, DWT-based image matching, and graphical PIN input. The Dashboard Module connects to the File class, which includes properties like file ID, filename, encryption key, and owner ID.

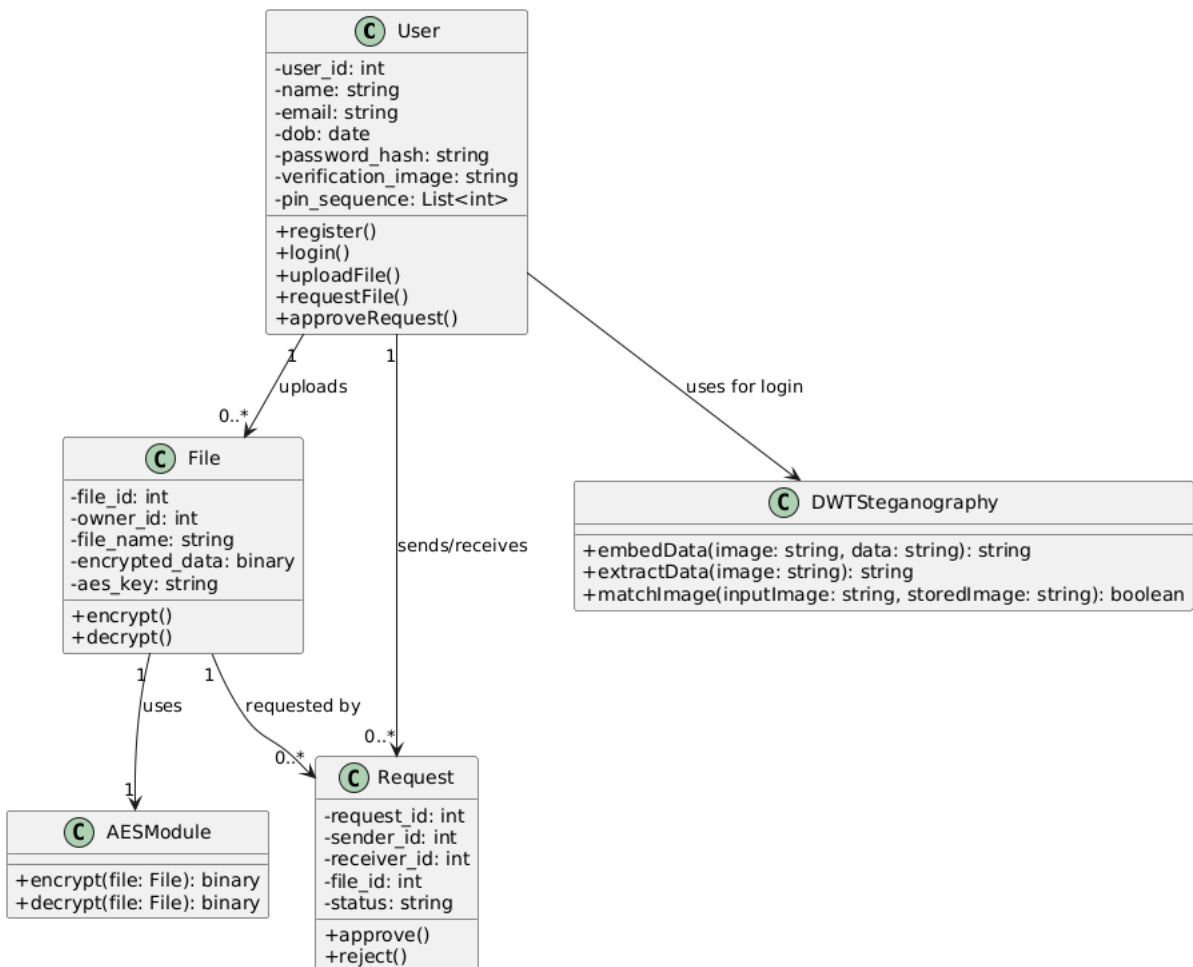


Fig 4.4 Class Diagram

A File Sharing Module class manages file access requests and approvals. The AES Module is responsible for encrypting and decrypting files, while the Steganography Module supports hiding and verifying image-based data. The Admin class has access to all system logs and user management methods, ensuring administrative control.

4.3.3 Activity Diagram

The Activity Diagram for the Graphical Password Authentication and Secured File Sharing System outlines the user journey from registration to secure file sharing. The process begins when a new user registers by submitting personal details, a password, a verification image, and selecting a graphical PIN. During login, the system verifies credentials through three sequential steps: email-password validation, DWT-based image verification, and graphical PIN input. Upon successful authentication, the user accesses the dashboard to upload files, which are encrypted using AES. Users can then request files from others or approve incoming file requests. If a file request is approved, the system decrypts the file for the authorized user. Failed authentication attempts redirect the user to retry the login process, maintaining security throughout.

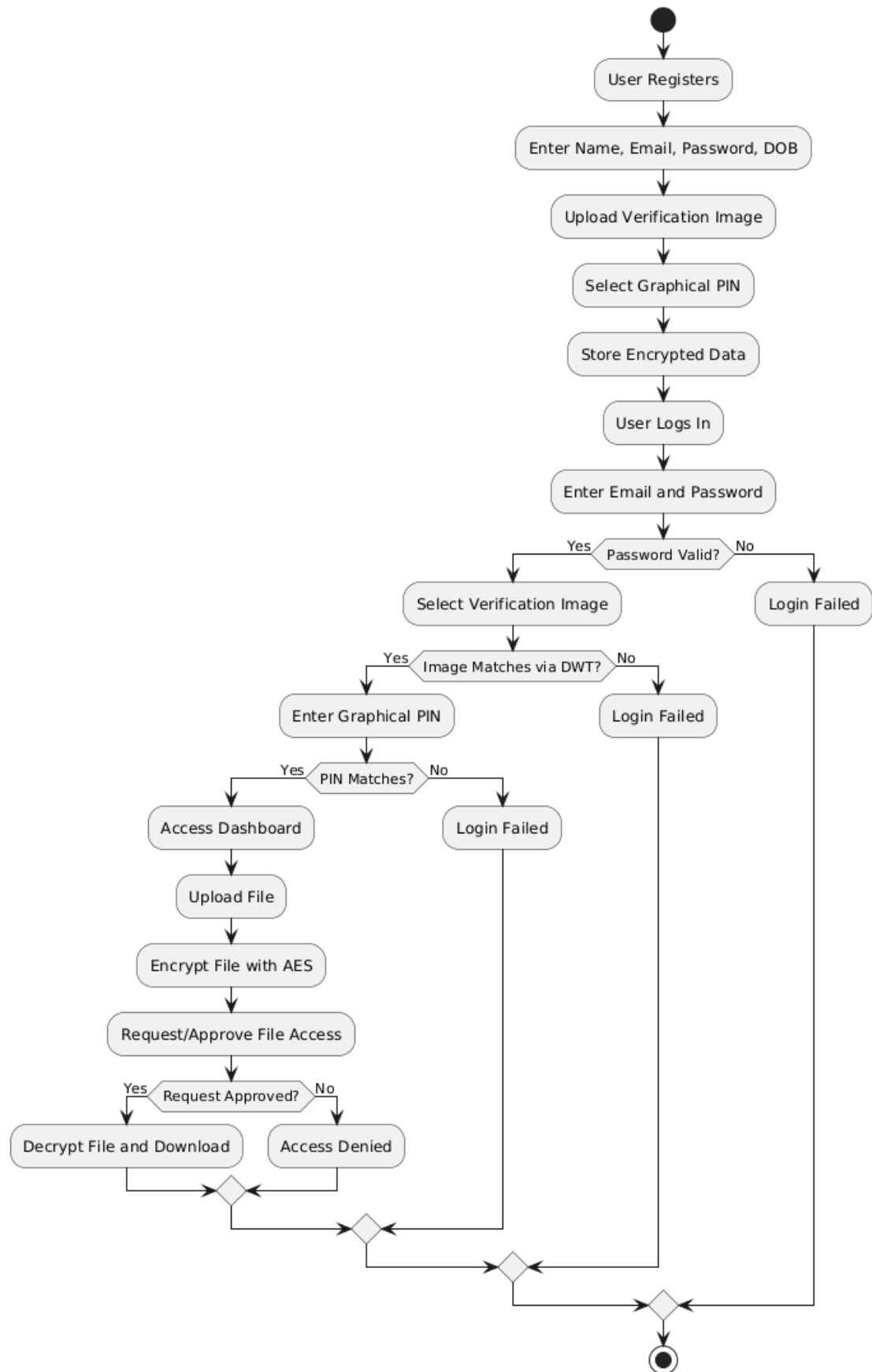


Fig 4.5 Activity Diagram

4.3.4 Sequence Diagram

The Sequence Diagram illustrates how the User interacts with the Authentication System and File Sharing System to securely access and exchange data. The process begins when a user initiates the login request by submitting their email and password to the Authentication System. Upon successful validation, the system prompts the user to select their pre-registered verification image. This image is matched using DWT-based steganography. If verified, the user is then asked to enter their graphical PIN. Once all three authentication layers are passed, the system grants access to the user dashboard. From the dashboard, the user can upload a file, which is then encrypted using the AES module and stored securely. If the user requests to access a file, the File Sharing System sends a notification to the file owner for approval. Once approved, the system automatically decrypts the file, making it available for secure download by the requester. This sequence ensures end-to-end authentication, encryption, and controlled access.

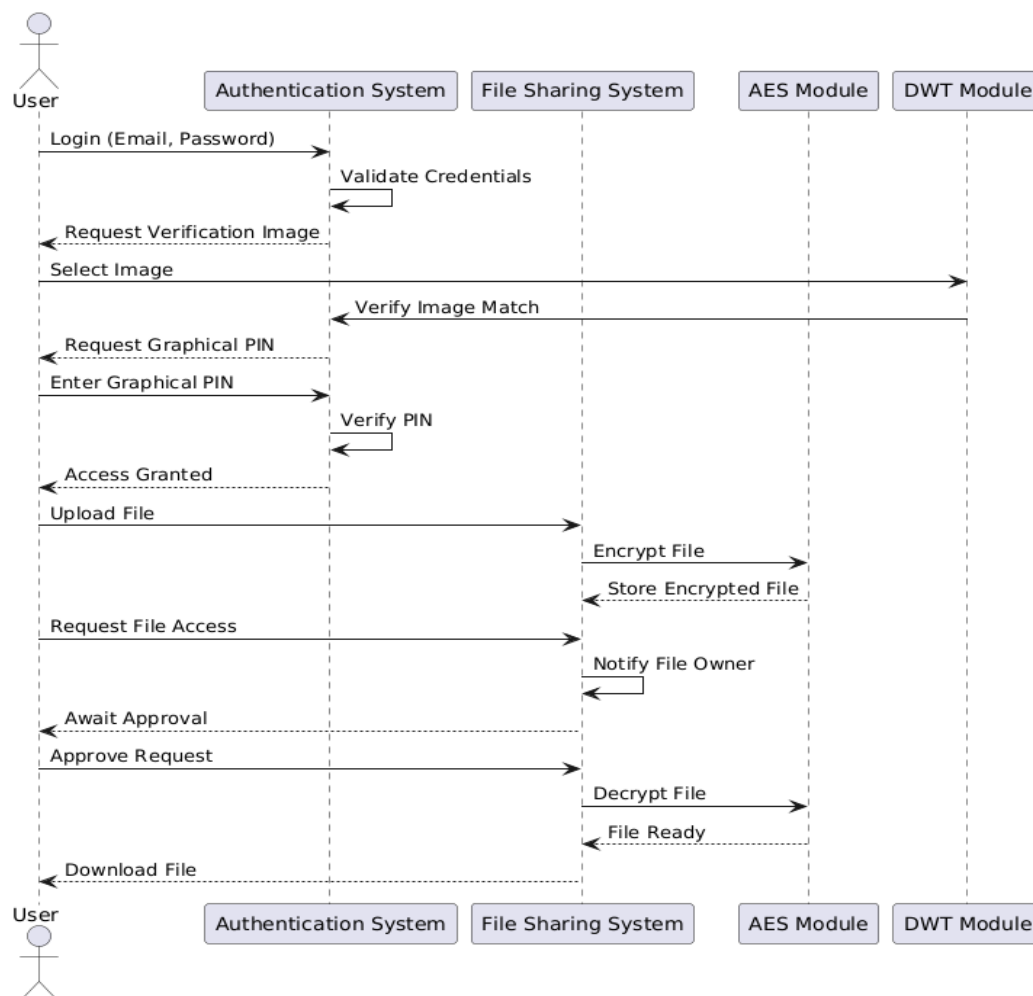


Fig 4.6 Sequence Diagram

CHAPTER 5

IMPLEMENTATION

5.1 MODULES

The Graphical Password Authentication and Secured File Sharing System consists of several essential modules that together ensure secure authentication, robust file encryption, and controlled file sharing between users. These modules work in unison to provide a highly secure and user-friendly platform.

The List of Modules are:

- User Registration Module
- Graphical Password Authentication Module
- Dashboard Module
- File Sharing Module
- AES Encryption Module
- DWT-based Steganography Module
- Security and Privacy Module
- Admin Module

5.2 MODULES DESCRIPTION

5.2.1 User Registration Module

Purpose:

This module allows new users to register for the system by providing their personal information and creating authentication credentials. It is designed to collect essential data from the user to initiate the authentication process.

Key Features:

- **Personal Details Collection:** The user provides name, email, date of birth, and password.
- **Verification Image Selection:** Users are prompted to select a verification image, which will be used for DWT-based steganography for secure authentication.

PIN Selection: Users select a 4-digit PIN by associating numbers with specific images. These images form the basis of the graphical password mechanism.

Data Storage: All details, including the encrypted password and PIN-image associations, are securely stored in the database.

Technologies: HTML, CSS, Flask (Backend), SQLite (Database), DWT-based steganography

5.2.2 Graphical Password Authentication Module

Purpose:

This module is responsible for the multi-layered authentication process during user login. It uses a combination of traditional credentials and image-based authentication to ensure security.

Key Features:

- **Email and Password Verification:** The system first checks the user's credentials (email and password).
- **Verification Image Matching:** Using the DWT-based method, the system ensures the correct verification image is selected.
- **PIN Pattern Matching:** The user is asked to reproduce the PIN pattern by selecting images corresponding to their original PIN setup.

Login Success: If all layers are passed, the user is granted access to the dashboard.

Technologies: Flask (Backend), OpenCV (for image processing), DWT-based Steganography

5.2.3 Dashboard Module

Purpose:

The dashboard module provides a user interface where users can interact with the system once authenticated. It includes file upload and file sharing features.

Key Features:

- **User Profile Management:** Users can view and update their personal information.

File Upload: Users can upload files (images, documents, or zip folders) to the system. The files are encrypted using AES before storage.

File Sharing: Users can request files from others, and share their own files after approval.

Encryption and Decryption: Files uploaded are encrypted using the AES algorithm. The recipient can download and decrypt the file securely.

File Management: The module allows users to view and manage uploaded and received files.

5.2.4 File Sharing Module

Purpose:

This module handles secure file sharing between users. It ensures that only authorized users can access the shared files and that all transferred data remains confidential through encryption.

Key Features:

- **Request and Approve File Sharing:** Users can request files uploaded by others. File owners must approve or deny the requests.

File Download and Decryption: Once approved, the recipient can securely download the file, which is automatically decrypted upon access.

Audit Logs: A log of all file requests and downloads is maintained to ensure transparency and security.

Technologies: Flask (Backend), AES Encryption, Secure File Transfer Protocols

5.2.5 AES Encryption Module

Purpose:

This module is responsible for encrypting and decrypting files using the AES algorithm. It ensures that files uploaded by users are stored securely and remain confidential during file sharing.

Key Features:

File Encryption: When a user uploads a file, it is encrypted using AES before being stored on the server.

File Decryption: When a user downloads a file, it is automatically decrypted using the AES algorithm.

Key Management: The AES keys are securely stored and managed to prevent unauthorized decryption.

Technologies: AES Encryption Algorithm (Python Cryptography)

5.2.6 DWT-based Steganography Module

Purpose:

This module uses Discrete Wavelet Transform (DWT) to embed authentication information in the user's verification image. It enhances the security of the graphical password system by hiding sensitive data within an image in a way that is hard to detect or manipulate.

Key Features:

Image Embedding: Authentication features (such as a user's ID or PIN) are embedded within a selected image using DWT.

Steganography Retrieval: During login, the system retrieves and matches the hidden authentication data to verify the user's identity.

Data Integrity: Ensures that any alteration to the image is detected, maintaining the integrity of the authentication process.

Technologies: OpenCV, DWT (Discrete Wavelet Transform)

5.2.7 Security and Privacy Module

Purpose:

This module ensures that all aspects of the system are secure and protect user data from unauthorized access or tampering. It includes mechanisms for secure storage, encrypted communications, and overall privacy protection.

Key Features:

Secure Storage: Sensitive user data, including passwords and PINs, are stored in an encrypted form.

HTTPS Encryption: All communication between the client and server is encrypted using HTTPS to prevent data interception.

Audit and Logging: The system maintains a detailed log of login attempts, file access, and modifications to ensure accountability and traceability.

User Privacy: The system ensures that users' personal information is never exposed during authentication or file sharing.

Technologies: HTTPS, Flask Security, Python Cryptography, JWT (for session management)

5.2.8 Admin Module

Purpose:

The admin module is for system administrators to manage user accounts, monitor system activity, and ensure the integrity of the authentication and file sharing processes.

Key Features:

User Management: Admins can view, create, and deactivate user accounts.

File Activity Monitoring: Admins can monitor file uploads, requests, and downloads.

System Audit: A comprehensive audit system allows admins to track every action taken by users and ensure compliance with security protocols.

Report Generation: Admins can generate reports regarding system activity and file sharing history.

Technologies: Flask (Backend), SQLite (Database), Admin Interface

CHAPTER 6

TESTING

6.1 TESTING AND VALIDATION

Testing plays a crucial role in ensuring that the Graphical Password authentication and Secured File Sharing System functions as intended and provides a robust defense against unauthorized access. The objective of testing is twofold: first, to uncover potential vulnerabilities or functional issues; and second, to resolve them to uphold system integrity and user trust.

During the validation process, the system undergoes rigorous testing across its core modules, including authentication, file encryption, file sharing, and image-based verification. Each module is tested individually (unit testing) and in combination (integration testing) to evaluate the overall system behavior under various use-case scenarios.

Unit Testing

Unit testing is conducted to evaluate each functional module of the system in isolation. This ensures that individual components operate correctly before system-wide integration. The following modules are tested:

- **User Registration:** Tests validate that users can register with valid details including name, email, date of birth, password, verification image, and graphical PIN. Input validation and secure data storage (hashed passwords, DWT-embedded images) are verified.
- **User Authentication:** This multi-layer authentication is tested at all levels—email and password verification, image recognition using DWT-based matching, and 4-digit graphical PIN input. The system correctly denies access on invalid inputs and permits access only when all layers are passed.

- **Dashboard Access and File Upload:** Tests confirm that only authenticated users can access the dashboard. File uploads (images, documents, zip folders) are tested for successful AES encryption and secure storage.
- **File Request and Sharing Module:** Unit testing checks if users can request files and whether file owners can approve or reject requests. It also ensures that files are only decrypted and downloaded after approval.
- **AES Encryption and Decryption:** Tests verify the integrity and confidentiality of file data. Encrypted files are tested to ensure they cannot be accessed without the proper AES key, and automatic decryption works only for authorized users.
- **DWT-based Steganography:** The module is tested for embedding and retrieving authentication features from images. The system ensures any tampering or mismatch is detected, maintaining the integrity of verification images.
- **Admin Module:** Admin functionalities such as user management, system audits, and activity monitoring are tested to ensure that they perform reliably and support system-wide oversight.

Each unit test confirms that respective modules perform as expected, handle invalid inputs securely, and integrate well during overall system operations.

Functional Testing

Functional testing ensures that the system performs all its intended operations accurately under various user scenarios. Key functionalities are tested to validate system reliability, usability, and security:

- **User Registration and Multi-Layer Authentication:** The system is tested to confirm that new users can register successfully, including selecting a verification image and graphical PIN. The authentication workflow is verified to ensure each layer—email/password login, image verification via DWT, and graphical PIN input—works as intended and prevents unauthorized access.

- **File Upload and Encryption:** Functional tests validate that users can upload different file types (documents, images, zip folders), and that each file is securely encrypted using the AES algorithm before being stored on the server.
- **File Request and Approval Workflow:** The system is tested for handling file-sharing workflows. This includes initiating file requests, approving or rejecting them, and confirming that file downloads only occur post-approval with successful decryption.

The following criteria were examined during functional testing:

- **Performance Test:** This test assesses whether login, file uploads, image verification, and file sharing occur without noticeable delays, ensuring a smooth user experience during authentication and data operations.
- **Stress Test:** Simulated high user loads were applied to evaluate system responsiveness and reliability when multiple users perform registration, login, file uploads, and file access simultaneously.
- **Structure Test:** Internal logic was validated for secure handling of DWT image verification, AES encryption/decryption routines, and proper coordination between user roles during file-sharing interactions.

Through functional testing, the system demonstrates robust handling of core tasks while maintaining user data security and system stability under operational conditions.

System Testing

System testing validates that all integrated modules of the Graphical Password Authentication and Secured File Sharing System work together as a cohesive unit. The objective is to ensure end-to-end functionality and reliability during real-world usage. The following components were tested:

- **User Authentication Workflow:** The full login process—from email and password validation, through DWT-based image verification, to graphical PIN

entry—was tested across all modules to confirm consistent session handling and secure user access.

- **Dashboard and File Management:** System tests validated the dashboard functionality, including file uploads, AES encryption, and organized access to uploaded and received files. The interaction between the authentication module and dashboard access control was also verified.
- **File Sharing and Access Control:** The file request and approval mechanism was tested for seamless communication between users. The system ensured that only approved recipients could download and automatically decrypt files, maintaining strict access permissions.
- **DWT Steganography and AES Integration:** The cooperation between the DWT steganographic verification and AES encryption modules was tested to confirm that image-based authentication and encrypted file operations could operate in parallel without conflict or error.

These tests confirm that the entire system functions smoothly as an integrated security platform, providing reliable multi-layered authentication and controlled, encrypted file sharing for end users.

Integration Testing

Integration testing is conducted to ensure that individual modules of the Graphical Password Authentication and Secured File Sharing System function correctly when combined. These tests confirm that the data flow and interactions between modules remain seamless and secure. Key integration scenarios include:

- **Authentication and Dashboard Access:** The integration between the multi-layer authentication module (email/password, DWT image verification, graphical PIN) and the dashboard is tested to ensure that only fully authenticated users gain access. The transition from login to dashboard must be smooth and secure.

- **File Upload and AES Encryption:** The connection between the dashboard module and the AES encryption system is tested to verify that all uploaded files are automatically encrypted before storage, and that metadata and encryption keys are correctly stored in the database.
- **File Request and Approval Workflow:** Integration tests ensure that file requests submitted by users are properly linked to the file owner's dashboard. Upon approval, the system must enable secure download and automatic decryption, confirming that all modules (file sharing, AES decryption, and access control) function cohesively.
- **DWT Steganography with Authentication Logic:** The integration between the image-based verification and DWT-based steganographic module is tested to validate that the hidden data is correctly matched during login and seamlessly supports the authentication process.

These integration tests confirm that the entire system operates as an interconnected and secure platform, offering robust authentication, secure file sharing, and a smooth user experience.

Acceptance Testing

Acceptance testing is carried out to confirm that the Graphical Password Authentication and Secured File Sharing System meets all functional and security requirements as defined in the project scope. The focus is on validating real-world usability, system accuracy, and user satisfaction. Key focus areas include:

- **Multi-Layered User Authentication:** The system is tested to ensure that users can complete the entire login process—email/password entry, image verification using DWT, and graphical PIN input—without confusion or failure, confirming that security layers are effective yet user-friendly.
- **Dashboard Usability and File Operations:** Users are able to access their dashboard post-authentication, navigate the interface smoothly, and perform

actions such as uploading encrypted files and managing personal data. The user interface is reviewed for clarity, responsiveness, and ease of interaction.

- **File Request and Sharing Process:** Acceptance testing verifies that file requests and approvals operate as intended. Users are able to send, receive, approve, and download files without technical issues, ensuring a seamless workflow from request to secure decryption.
- **Security Features and Privacy Compliance:** Tests confirm that sensitive data is encrypted and handled according to best practices. Steganographic verification and AES encryption work reliably, providing both security and confidence to end users.

These tests confirm that the system aligns with business expectations and provides a secure, accessible, and efficient platform for authentication and file sharing.

6.2 BUILD THE TEST PLAN

The testing strategy for the Graphical Password Authentication and Secured File Sharing System was structured into three primary stages—unit testing, functional testing, and system testing—to ensure modular reliability, smooth integration, and overall system robustness. The test plan included:

- **User Authentication:** Testing was conducted across all three authentication layers—email/password, DWT-based image verification, and graphical PIN input. Each layer was validated independently and in sequence to ensure secure and consistent user access.
- **Dashboard and File Management:** The dashboard module was tested to confirm that users can upload files, which are then encrypted using AES. File listings, updates, and user profile interactions were examined for accuracy and usability.
- **File Sharing and Request Handling:** Tests were performed to verify the functionality of the file request and approval system. The process of sending,

receiving, approving, and downloading files—along with auto-decryption—was thoroughly checked for correctness and security.

- **DWT and AES Integration:** Specialized tests validated that DWT-based steganographic verification and AES encryption/decryption modules function together without conflict, maintaining high performance and data integrity.
- **System Performance:** Performance and stress testing were carried out to assess the system’s ability to handle multiple simultaneous users, ensuring no degradation in authentication speed, file processing, or data transfer.

Each module was first tested in isolation before integration, ensuring that all individual components were self-contained and error-free. Once validated, modules were combined and retested to guarantee cohesive operation and a secure, seamless experience for all users.

Table 6.1 Test Case Design

Test Case ID	Test Description	Test Procedure	Test Input	Expected Result	Actual Result
TC001	To verify if user registration works correctly	Open the system, navigate to registration, fill required fields and submit	Name, Email, DOB, Password, Verification Image, Graphical PIN	System should register the user and store data securely	User registered successfully and redirected to login
TC002	To verify multi-layer authentication during login	Attempt login using correct credentials, select verification image, and enter correct graphical PIN	Email, Password, Image, Graphical PIN	System should authenticate the user and open dashboard	Login successful and dashboard accessed

Test Case ID	Test Description	Test Procedure	Test Input	Expected Result	Actual Result
TC003	To verify that file upload encrypts the file securely	Log in, navigate to dashboard, upload a file	File (e.g., image/document/zip)	File should be encrypted and stored securely	File uploaded and encrypted successfully
TC004	To verify file request feature	User A requests a file uploaded by User B	File ID	File request should be recorded and visible to the uploader	File request successfully created
TC005	To verify file sharing approval mechanism	User B logs in, views pending requests, and approves a file request	Request ID	Approval should allow the file to be decrypted by the requester	File shared and decrypted by recipient
TC006	To verify image-based DWT steganographic matching	Log in and select correct verification image	User-selected image	Image should match DWT signature for access	Verification image matched successfully
TC007	To verify graphical PIN input validation	During login, enter image sequence matching registered PIN	PIN image sequence	System should validate and allow access if correct	PIN verified and access granted
TC008	To verify file decryption upon approved download	After approval, recipient downloads a shared file	File ID and AES Key	File should be decrypted and downloaded securely	File decrypted and downloaded correctly

Table 6.2 Test Case Log

Test ID	Test Description	Test Status (Pass/Fail)
TC001	To verify if the system correctly prompts for login.	PASS
TC002	To verify if the system handles invalid login attempts gracefully.	PASS
TC003	To verify if the system matches the verification image using DWT-based steganography.	PASS
TC004	To verify if the system allows users to upload files with AES encryption.	PASS
TC005	To verify if the system successfully validates the graphical PIN pattern during login.	PASS
TC006	To verify if the system allows users to request access to shared files.	PASS
TC007	To verify if the file owner can approve or deny file requests.	PASS
TC008	To verify if the system automatically decrypts files upon approved download.	PASS
TC009	To verify if the system maintains audit logs for login attempts and file-sharing activities.	PASS
TC010	To verify if the system ensures HTTPS encryption for all communications.	PASS

CHAPTER 7

RESULT AND DISCUSSION

The evaluation of the Graphical Password Authentication and Secured File Sharing System confirms its effectiveness in providing a secure and user-friendly platform for authentication and file sharing. Through comprehensive testing, the system consistently delivered a seamless experience across all user interactions, including registration, multi-layered authentication, and file management.

Users were able to register, authenticate, and share files with ease, validating the system's usability and interface responsiveness. The multi-layered authentication approach—combining email-password verification, DWT-based image recognition, and graphical PIN patterns—significantly enhanced security while maintaining user convenience. This reduced the risk of unauthorized access and ensured a robust login process.

The AES encryption mechanism ensured the confidentiality and integrity of files during upload, storage, and sharing. The controlled file-sharing module, with its request-and-approval workflow, provided users with complete control over access permissions, further enhancing data security.

System performance was tested under both typical and edge-case scenarios. Authentication processes, dashboard interactions, and file sharing operated within optimal time frames. When handling edge cases—such as incorrect login attempts, mismatched PIN patterns, or unapproved file requests—the system displayed appropriate error messages and guided users through corrective actions. For instance, if a user attempted to access a file without approval, the system restricted access and notified the file owner, showcasing robust error handling and user-centric workflows.

The integration of DWT-based steganography for image verification ensured secure storage and retrieval of graphical authentication features. This

method provided an additional layer of protection against brute-force and phishing attacks while maintaining high accuracy in matching verification images.

The user dashboard proved to be highly functional, offering features like file upload, encrypted storage, and real-time file-sharing updates. The inclusion of audit logs ensured traceability of all activities, including login attempts and file-sharing actions, enhancing system transparency and accountability.

Overall, the combination of advanced image processing techniques, multi-layered authentication, and AES encryption elevated the platform's security and usability. This makes the system a reliable solution for secure authentication and data sharing in academic, corporate, and personal applications. The evaluation confirms its value as a modern and robust security framework for protecting sensitive information.

CHAPTER 8

USER MANUAL

Installing Required Tools and Libraries

Installing Python

- Step 1: Visit the official [Python Downloads page](#).
- Step 2: Choose the version compatible with your operating system (Windows, macOS, or Linux) and download the installer.
- Step 3: Follow the prompts to complete the installation.
- Step 4: Verify the installation by opening a terminal or command prompt and typing `python --version`.

Installing Flask

- Step 1: Open a terminal or command prompt.
- Step 2: Create a virtual environment using `python -m venv venv`.
- Step 3: Activate the virtual environment:
 - On Windows: `venv\Scripts\activate`
 - On macOS/Linux: `source venv/bin/activate`
- Step 4: Install Flask by running `pip install flask`.
- Step 5: Verify the installation by typing `flask --version`.

Installing SQLite

- Step 1: Visit the SQLite Downloads page.
- Step 2: Download the precompiled binary for your operating system.
- Step 3: Follow the instructions provided to set up SQLite on your system.
- Step 4: Test the installation by typing `sqlite3` in the terminal.

Installing Visual Studio Code (VS Code):

Step 1: Visit the Visual Studio Code Downloads Page.

Step 2: Download the VS Code installer for your operating system (Windows, macOS, or Linux).

Step 3: Run the installer, follow the setup prompts, and accept the license agreement to complete the installation.

Step 4: Launch Visual Studio Code and install recommended extensions for your project, such as the “Prettier” and “ESLint” extensions for code formatting and linting.

Running Tests in VS Code:

Step 1: Open the terminal in VS Code and navigate to your project directory.

Step 2: To execute unit tests, run `npm test` for the backend and `npm run test` for the frontend, if configured.

Step 3: Review the test output in the terminal to identify any errors or issues.

CHAPTER 9

CONCLUSION

The Graphical Password Authentication and Secured File Sharing System successfully provides a secure, efficient, and user-friendly solution for authentication and data sharing. Its multi-layered security approach—combining traditional login credentials, DWT-based steganography for image verification, and graphical PIN authentication—ensures robust protection against common security threats, such as brute force, phishing, and shoulder surfing attacks. The inclusion of AES encryption further enhances the system’s ability to safeguard sensitive files during upload, storage, and sharing. The system’s seamless integration of advanced image processing techniques and encryption methods offers a comprehensive framework for secure operations. It demonstrates resilience by handling both typical operations and edge cases with strong error management and user-centric workflows. For example, the system effectively guides users through corrective actions when login or file-sharing prerequisites are not met. The intuitive user interface simplifies interactions, while the controlled file-sharing mechanism ensures that data access remains transparent and secure. The system's modular design supports scalability and future enhancements, such as integrating biometric authentication or cloud-based deployment, allowing it to adapt to evolving technologies and user needs. In conclusion, the Graphical Password Authentication and Secured File Sharing System sets a modern standard for secure authentication and data sharing platforms. By combining security, usability, and scalability, it highlights how innovative approaches can improve user privacy, enhance data protection, and streamline workflows in academic, corporate, and personal contexts.

CHAPTER 10

FUTURE ENHANCEMENT

The Graphical Password Authentication and Secure File Sharing System aims to evolve with advanced features that enhance security, scalability, and user experience. Future updates may include biometric authentication (e.g., fingerprint or facial recognition), multi-language support, and cloud-based deployment for global accessibility. Enhanced file-sharing controls—such as expiry-based access, watermarking, batch uploads, and restricted downloads—will offer greater user control. Automation of audit logs, data backups, and system alerts will boost efficiency and reliability. Integration with third-party tools (e.g., document editors, cloud storage) and AI-powered assistance will streamline user interaction. Real-time threat detection using AI and proactive defense mechanisms will strengthen cybersecurity. Support for decentralized storage (e.g., IPFS), blockchain-based audit trails, and adaptive user interfaces will further modernize the platform. Features like multi-user file collaboration and behavior-based UI personalization will make the system more interactive and secure. These enhancements ensure the platform remains future-ready, secure, and aligned with the dynamic needs of digital data protection.

APPENDIX-I

BASEPAPER

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 RESEARCH ARTICLE

SMFA: Strengthening Multi-Factor Authentication With Steganography for Enhanced Security

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ABSTRACT Traditional password-based authentication mechanisms exhibit significant security vulnerabilities, prompting the adoption of Multi-Factor Authentication (MFA) to enhance user verification. However, current MFA methods, such as OffPAD, One-Time Password Tokens (OTT), and smartcard-based systems, remain vulnerable to attacks like Man-in-the-Middle (MITM), session hijacking, replay, phishing, and denial of service (DoS). Additionally, the conventional single-server authentication approach is inefficient and inadequate for modern security needs. This study introduces Secure Multi-Factor Authentication (SMFA), a novel framework designed to address these challenges by leveraging steganography for secure credential transmission, reducing susceptibility to MITM and session hijacking attacks. SMFA also incorporates an additional Universal Serial Bus (USB) device as a supplementary authentication factor and employs a multi-server architecture to overcome the limitations of single-server models. A comprehensive analysis comparing SMFA to existing MFA protocols reveals its superior ability to counter MITM, replay, DoS, user impersonation, and password-guessing attacks. The results demonstrate that SMFA offers a robust and effective solution, addressing the weaknesses of both traditional password-based systems and current MFA methods.

INDEX TERMS User authentication, multi-factor authentication model (MFA), steganography, cryptography, privacy and security.

I. INTRODUCTION

Authentication plays a pivotal role in electronic systems by establishing the user's authenticity. This process involves the user proving their identity through three fundamental factors:

- 1) Something the user knows: This encompasses elements like the username, password, or specific questions directly related to the user, with answers known to the system.
- 2) Something the user has: This includes various items such as OTP, smartcards, NFC, and USB devices.

- 3) Something the user is: This refers to the user's biometric information, such as fingerprint, retina, and DNA, providing unique identifiers.

Collectively, these factors used to verify a user's identity are referred to as credentials. Despite the prevailing use of single-factor authentication, primarily based on usernames and passwords, this mechanism has raised security concerns since the late '70s [1] [2]. The primary vulnerability of the single-factor password-based system lies in password cracking through brute force and the risk of a user registering with the same password across multiple systems. Passwords are susceptible to compromise, as evidenced by numerous data breaches in recent years [3], [4], [5], [6], [7].

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A notable breach in late 2018 exposed the data of 162 million users of a popular video messaging service [8]. Additionally, significant data leaks occurred in 2019, including 530 million user records, including Face ID, from Facebook [9], and 885 million user records from the First American Financial Corporation [27], [10]. These incidents underscore the vulnerability of passwords in safeguarding sensitive information.

In addition to Facebook, recent cyber-attacks have targeted other social media platforms like LinkedIn and Sina Weibo, compromising a total of 1.2 billion records [11], [12]. Data breaches occur frequently due to inadequate security measures and policies. For instance, in 2022, a leak exposed data from 3 million patients, including SSNs, medical records, and sensitive information [13]. Given the security vulnerabilities inherent in traditional password-based systems, this research stems from the necessity of a robust authentication system.

Multi-factor authentication protocols combine numerous authentication elements to mitigate password flaws. Notably, FIDO's U2F [14] and Google 2-step [15] are two well-known multi-factor (mostly two-factor) authentication methods that have gained popularity. In particular, U2F makes use of a USB-based authentication system. These protocols are still vulnerable to frequent attacks like DOS and MITM, though. Additionally, various other authentication methods have emerged, offering specific advantages against types of attacks and security issues. Therefore, the primary concerns revolve around (i) verifying that the individual seeking access is the legitimate user and (ii) safeguarding the credentials. Thus, safeguarding user credentials during transmission over the network poses a significant challenge.

Securing the transmission of credentials plays a pivotal role in authentication, where cryptography and steganography stand out as crucial techniques. Steganography, an age-old method involving concealing data within a cover, effectively shields sensitive information [16]. In the digital realm, it utilizes images, videos, or audio files as cover objects [17]. While cryptography encrypts messages, steganography focuses on hiding their existence. Employing both ensures robust data security. Despite potential implementation expenses, industries such as banking and military networks favor steganography due to its high-security standards, making it a preferred choice for ensuring data security in authentication processes [18].

Previous studies have shown that adding a steganography system to the authentication procedure improves data security when credentials are being transmitted. Therefore, this study conducts a thorough investigation into incorporating steganography and cryptography into a newly structured authentication model. The aim is to address the security concerns stemming from conventional password-based systems. This paper contributes to the field in the following ways:

- 1) We carry out a thorough analysis of the entirety of research, highlighting critical weaknesses such as insider threats, user impersonation, Man-in-the-Middle (MITM) attacks, and session key leakage. We present a

multi-factor authentication paradigm that is intended to protect sensitive data that belongs to authorized users in order to address the security flaws in the protocols now in use.

- 2) We highlight the effective application of steganography to mitigate risks associated with critical attacks.
- 3) Burrows-Abadi-Needham (BAN) logic is used to assess the suggested authentication procedure in order to make sure it functions properly.
- 4) A thorough study backed up with declarations shows that the suggested protocol is secure from replay attacks, Man-in-the-Middle attacks, and other significant risks.
- 5) We undertake a comparison analysis, comparing the suggested protocol's security functionality and performance to previous research.

The structure of the paper is as follows: Section II delves into previous research in the relevant field. Section III outlines the proposed Multi-Factor Authentication (MFA) model, while Section IV encompasses discussion, analysis, and comparative results. Finally, Section V concludes the paper.

II. LITERATURE REVIEW

The initial phase of any secure and dependable transaction involves authentication. Conventional password-based methods play a predominant role in this crucial step to verify the user's authenticity. Simultaneously, advanced authentication mechanisms are gaining widespread adoption worldwide.

During this research, it was noted that previous investigations have explored steganography approaches and various types of multi-factor authentication mechanisms. The following section provides an overview of the relevant work conducted in this domain.

A. MFA MODELS AND PROTOCOLS

Numerous prior research endeavors have explored multi-factor authentication methodologies to verify user identities. For distributed wireless sensor networks, Das devised a three-factor authentication mechanism, claiming its resilience against certain common attacks [19].

Subsequently, Babu et al. scrutinized Das's protocol and identified its vulnerability to replay attacks and session-specific temporary information threats. An enhanced protocol addressing these shortcomings was proposed [20]. In another study [21], an authentication protocol for multi-server environments utilizing a two-factor system (password and smart-card) with RSA cryptography was presented. This protocol aimed to streamline complexities while fortifying security against known attacks. Validation showcased its effectiveness in mitigating recognized security threats. Additionally, the protocol facilitated session key agreement and mutual authentication, verified for correctness and session key freshness using the BAN logic model. Another work [22] introduced a robust three-factor mutual authentication protocol tailored for e-governance systems operating within multi-server environments. Its security was

substantiated through a comprehensive assessment involving informal security analysis, BAN logic, ROR model, and simulation via the AVISPA tool.

Shen J. et al. proposed a lightweight authentication system for cloud environments, leveraging XOR, string concatenation, and a hash function to reduce computational overhead on both client and server sides [23]. Meanwhile, Yang et al. introduced an ID-based user authentication, an enhancement of an existing scheme, and another authentication scheme for multi-server environments. However, these were designed for cloud settings, slightly escalating computational costs [24]. A concrete instance of an identity-based authentication technique was provided by Wang et al. and Das et al. [25], [26]. Although data transmission is not safe in the Wang et al. model, both models were suggested for correct user authentication. It is discovered that credentials are modifiable during transmission across the network in this regard. Das et al.'s methodology is likewise insecure since it allows for authentication bypassing without requiring any legitimate credentials. To get around the one-pass authentication process, Liu suggested the one-pass authentication and key agreement (AKA) approach [27]. Man-in-the-middle, session hijacking, and server spoofing attacks are just a some of the potential threats that Huang and Li discovered when they examined the AKA [28]. However, Tsay and Mjølunes discovered a flaw that was open to internal and external attack [29].

B. MFA USING ADDITIONAL DEVICE

Over the past decade, numerous sophisticated methods have emerged, advocating the inclusion of an additional device to verify user authenticity. These approaches have prompted research studies advocating for the integration of smartcards, USBs, and supplementary online devices as an extra authentication factor. A multi-factor authentication system using a user ID, password, and smartcard was proposed by Choudhury et al. [30]. Nevertheless, smartcards are not appropriate for public cloud systems and have limited usability.

An MFA system using IMEI and OTP numbers as authentication secrets in mobile contexts was introduced by Kumar et al. [31]. The concept of OffPAD, initially introduced by Varnedal et al. [32] and detailed in 2005, aimed to manage and authenticate both service providers and user identities, particularly for online transactions to prevent MITM and phishing attacks. Subsequently, Alhaidary et al. conducted vulnerability analyses and proposed enhancements to address these issues [33]. In 2016, their investigation highlighted vulnerabilities in the OffPAD-based authentication model, suggesting techniques to mitigate susceptibilities [34], revealing its extreme vulnerability to replay and MITM attacks.

Bae and Kwak proposed an authentication scheme using smartcards within a multi-gateway IoT framework to reduce communication and computational overhead [35]. However, their scheme lacked resilience against several security

threats, including traceability, impersonation, gateway node spoofing, and session key disclosure attacks, while also failing to provide secure mutual authentication.

Khan et al. developed a secure mutual authentication system for smart grid communication, employing biometric-based elliptical curve cryptography, ensuring user anonymity, defending against various attacks, and reducing transmission and computation costs [36]. In response to security challenges, Masud et al. devised a user authentication mechanism prioritizing lightweight implementation and user anonymity preservation [37].

Widely used Advanced Encryption Standard (AES) was used for data security in Hufstetler et al.'s implementation of a safe multi-factor authentication scheme that replaces authentication in the current Windows version through NFC [38]. However, limitations confined its use to Windows PCs, and security threats persisted in the authentication process.

To address security concerns and enhance functionalities, three-factor authentication protocols utilizing biometric information were proposed. Although user anonymity was not guaranteed, Liu et al. proposed a biometric-based single sign-on authentication mechanism for mobile cloud computing services [39]. Chen and Chen also suggested using three factors for authentication in e-health cloud systems, but the systems were still vulnerable to intractability and session-specific temporary information attacks [40], [41].

C. MFA USING STEGANOGRAPHY IN WEB

Authentication protocols play a pivotal role in facilitating secure data exchange and communication among system entities by leveraging cryptography [42]. According to Liu, these protocols primarily ensure key agreements, data security, non-repudiation, and multi-party computation [27]. Steganography has been explored in various research studies within web authentication, presenting an opportunity to enhance authentication mechanisms through its application. Combining steganography with cryptography offers a promising avenue to bolster data security and conceal sensitive credentials.

Studies have proposed combining hashing or encryption with steganography to accomplish strong data security in digital communications. Madhuravani et al. introduced a method utilizing dynamic hashing and steganography [43], highlighting the effectiveness of combining steganography with cryptography to deceive attackers. An authentication framework called imgAuth that uses picture steganography for user profile management and authentication was described by Gunawardena et al. [44].

Choudhury et al. proposed a method employing steganography and visual cryptography for secure information sharing between customers and merchant servers, prioritizing enhanced safety [30]. Tabassum et al. [45] showcased an improved version of the Kerberos protocol, integrating biometric templates and steganography to heighten security, while Motero et al. conducted a comprehensive analysis, demonstrating its exemplary effectiveness [46].

A safe authentication method for online purchasing systems was presented by Ihmaidi et al. [47]. This mechanism uses steganography and biometrics to hide login credentials. This method uses a particular steganography algorithm to encrypt credit cards and login passwords, putting them inside an image. Mantoro et al. described in detail how to use steganography techniques to hide text passwords within graphics [48]. Furthermore, in order to guarantee data security, Danuputri et al. introduced a prototype known as ste-Chy that combines the Vigenère cipher and LSB steganography for private authentication [49]. SHA-256 was applied in order to guarantee legitimacy. The study asserts that using steganography for online authentication greatly improves data security, although it focuses on the Android environment in particular.

Hence, it is evident that numerous initiatives and models have been put forth to enhance the security of authentication processes. Multi-Factor Authentication (MFA) emerges as a superior choice, providing a heightened security level compared to the single-factor authentication system. Additionally, the potential application of steganography in authentication is apparent, but it necessitates further examination. Previous research has separately delved into steganography and authentication models.

However, little research has been done on the combined use of steganography with multi-factor authentication. Steganography has been utilized in the e-commerce industry to secure credential transfer, and it has also been integrated by certain researchers into cloud authentication models. This work makes a contribution by developing a multi-factor authentication mechanism that transmits credentials securely by using steganography. This study focuses on the smooth integration of steganography with the authentication system, taking into account the following aspects when designing the model:

- 1) Employing cryptography for robust data protection.
- 2) When the transmitted packet contains several possible cover objects, steganography is used to hide the presence of credentials within a cover item. This method works well for deceiving attackers and extending the amount of time needed to decode secret credentials.
- 3) Designing the model to allow the use of two distinct servers to mitigate risks and enhance efficiency.

III. RESEARCH METHODOLOGY

Multi-factor authentication surpasses single-factor authentication methods in terms of providing enhanced security. Various organizations already implement specific MFA protocols or models. The subsequent discussion revolves around these existing MFA mechanisms.

A. BACKGROUND

1) OFFPAD

OffPAD serves as an offline personal authentication device specifically designed to empower users in securely managing online transaction authentication processes. Its primary

objective is to oversee online identification procedures, effectively preventing MITM and phishing attacks. Initially detailed by Josang and Pope in 2005, this personal authentication device was subsequently improved upon by Vannedal et al., leading to the development of OffPAD [32]. The device's unique feature lies in its offline nature, predominantly operating with a secure element to ensure heightened security.

2) REMOTE USER AUTHENTICATION SCHEME USING SMART CARDS

Using ElGamal's public key cryptosystem, Hwang and Li suggest integrating a smart card into user authentication [50]. This system's security depends on how difficult it is to compute discrete logarithms in finite fields. It operates as a remote user authentication system without employing a password verification table. During registration, users link their smartcard with a password. Subsequently, an authorized user seeking system access must insert the card into the device along with their ID and password during login attempts.

3) GOOGLE 2-STEP

Google introduces a two-factor authentication method called Google 2-step [15]. Primarily phone-based, this scheme allows users to utilize their phones to validate logins. Google highlights on their website several reasons why relying solely on passwords for authentication isn't adequate, asserting that '2-Step Verification can help keep bad actors out, even if they possess your password'.

During login, a verification code is dispatched to the user's computer, which needs to be entered for access. An alternative version, the 'one-tap' variant, simplifies the process by prompting users to press a 'yes' button in a phone pop-up. This second version notably enhances usability compared to its predecessor.

4) FIDO

There is a FIDO authentication standard variant that uses the U2F USB token as the second factor [14]. In addition to storing secrets, this U2F USB token may also include public keys. These keys are essential for carrying out cryptographic procedures. The USB token is notable for having a button that users must press in order to validate a transaction. On its website, it states that the public key is registered on the server, and the token creates a key pair that allows a website to enable second-factor authentication. The token can be used for authentication once this one-time procedure is complete.

Through this method, the computer communicates with the server the user's login credentials and password. The server then creates a challenge, which is sent back to the user's PC. After that, the browser creates a payload that includes the challenge, the server's URL, and the session that needs to be signed by the token.

B. PROPOSED AUTHENTICATION MODEL

A thorough explanation of the suggested MFA model, which may be applied to a dependable and safe authentication

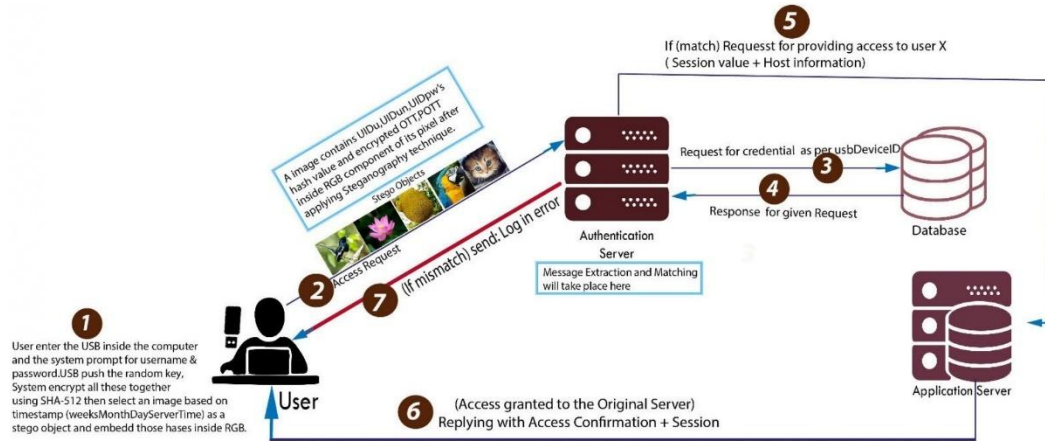


FIGURE 1. Proposed SMFA model.

procedure, is provided in this section. The suggested approach hides the existence of data and ensures data security using steganography and cryptography.

The proposed mechanism introduces a USB device as an additional authentication factor alongside existing credentials like usernames and passwords. Operating under the assumption of three distinct roles within this system—namely, the user, the authentication server, and the application server—specific functions are assigned to each role. The application server serves authorized users, while the authentication server verifies user authenticity. During login, users are prompted to utilize a registered USB device to confirm their identity.

As shown in Figure 1, this technique consists of three main phases: the request phase, the authentication phase, and the response phase. During the server login request procedure, users are required to utilize their registered USB device, which has a Stego validator that generates and stores One-Time Tokens (OTTs).

Assuming User A tries to log in, the registration procedure is skipped for the sake of this study. As such, the registration phase is skipped in this context and the proposed USB-based, multi-server authentication system consists of three main phases: the login phase, the authentication phase, and the registration phase. Whenever a new user signs up on the server or within the system. The authentication server provides the user with a USB device to produce the one-time token after receiving the registration request. Below is a description of each step.

1) THE REQUEST PHASE

- Login is requested by User to the system (S_a).
- User A enters USB device to the system (S_a). System reads UID_u and the banner Bo enclosed within the Stego validator S_{vu} .

- Bo is matched by Client side. If match found, then system prompts the user to provide their UID_{un} and UID_{pw} .
- Upon prompted by the system, User A provides UID_{un} and UID_{pw} (UID_{un} , UID_{pw}).
- Newly generated OTT from S_{vu} (UID_{un} , UID_{pw}), timestamp Tsc is received by client side.
- Hash operation is performed on the following UID_u , UID_{un} , utilizing the SHA-512 hashing algorithm at the client side. Thereafter, employing asymmetric algorithm, encryption on OTT and POTT (OTT , $POTT$) is performed.
- One image is chosen from a collection of images (group of 3/4/5/7/10) based on the OTT stored in S_{vu} .
- Use the appropriate and efficient steganography technology to embed the hashed data inside the chosen image.
- Send this image together with other images after the credentials have been embedded and encrypted $E_n(OTT, POTT)$ to the authentication server S_p .

2) THE AUTHENTICATION PHASE

- S_a obtains the picture files and $E_n(OTT, POTT)$.
- After successfully decrypting the $E_n(OTT, POTT)$, find the STo_i and pull out HASH, $H(h1)$ from STo_i .
- Execute the hashing process on $H(h1)$ and produce $H(h2)$.
- $Rsh = h2$ — Ch and Rott = POTT — DBOTT
- Store OTT in DBOTT only if Rsh and Rott both is TRUE. Redirect to S_a .

3) THE RESPONSE PHASE

- Access request to C_u is received by S_p .
- A session is created by S_p .
- Finally, access is given to C_u .

TABLE 1. Proposed authentication model notation.

Notation	Description
S_a	Authentication server
S_p	Application server
S_{vu}	Stego validator
UID_u	USB identification number
UID_{un}	Username
UID_{pw}	Password
$ID_{r,u,a}$	Authentication server identity request
OTT	Stego validator generated One-time token
POTT	Previously used OTT stored in stego validator
$H(h)$	Secure one-way hash function
$\parallel$	Compare Operator
E_n	Encrypted

4) THE RECOVERY PHASE

In the scenario of USB device loss or damage, this research presents two suggested approaches. Firstly, users are advised to register an additional USB device to mitigate the impact of such sporadic incidents. Registering multiple devices enables users to have at least two USB sticks registered, allowing access with the other device in case of loss or damage to one. The reporting process should be user-friendly, while the authentication server needs to revoke the stego validator and OTT of the lost USB device upon confirming the authenticity of the loss or breakage report.

This study suggests that using multiple registered USB devices is the best course of action since it reduces the need and bother of account recovery. This strategy, meanwhile, might not be totally adequate. A re-registration procedure may be taken into consideration as a solution to this limitation, subject to the organization's unique user situation, security requirements, and regulations. Reissuing account details and verifying one's identification in full, however, can result in higher service overhead. It's important to note that the SMFA procedure doesn't explicitly advocate for this but presents it as a potential alternate mechanism. The decision on which approach to adopt should be made by organizations, taking into account risk management and security policies. Figure 1. illustrates the architecture of the proposed model, which integrates the SHA-512 HASH function and RSA encryption to conceal original data.

This model leverages multiple images, selecting a single image to conceal data using an appropriate steganography technique. POTT manages the image selection for steganography application. There are other data-hiding methods available in the discipline of image steganography, but the most popular and straightforward one is the LSB replacement algorithm. Any image steganography method with a strong security level, however, can be used with this framework. The peak signal-to-noise ratio (PSNR) and mean square error (MSE) between the Cover object and the Stego object are taken into account during the security evaluation process. The following formula can be used to get the PSNR [51] between

two images:

$$MSE = \frac{1}{mn} \sum_{i=1}^m \sum_{j=1}^n (x_{ij} - y_{ij})^2$$

m number of rows in cover image
 n number of columns in cover image
 x_{ij} pixel value from cover image
 y_{ij} pixel value from stego image

$$PSNR(x, y) = \frac{10 \log_{10} [\max(\max(x), \max(y))]^2}{x - y^2}$$

The pixel values at places i and j in the Cover picture are represented by the equations as x_{ij} , and the pixel values at positions i and j in the Stego image are indicated by y_{ij} . Research indicates that when the PSNR (peak signal-to-noise ratio) between two images (Cover image and Stego image) exceeds 40dB, it indicates good quality. A higher PSNR implies lower imperceptibility. Notably, between two identical and unchanged pictures, the PSNR is infinite. Hence, any image steganography technique achieving a minimum of 40dB PSNR in this model suffices to conceal credentials effectively.

The recommendation is to employ an algorithm yielding the maximum PSNR to minimize perceptibility, signifying a higher level of security. The general procedure for concealing credentials in this model follows these steps:

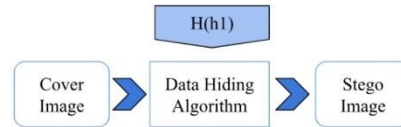


FIGURE 2. Procedure of cover object generation.

IV. MODEL ANALYSIS

This model uses an authentication server and an application server to create a shared key between three principals and a database. Although it uses the timestamp as a nonce, it is based on the shared key. Three principals- U, AU, and AP is provide for the above-mentioned proposed model. Thereafter, K_U , K_{AU} , and K_{AP} are their public keys; K_{UAU} , K_{UAP} , and K_{AUPU} are their shared keys; and DB is the authentication server database. Additionally, AU generates the timestamp T_{AU} , while U generates T_U ; and AT generates the timestamp T_{AP} . Moreover, the lifetime L is generated by U. Only in cases where mutual authentication is necessary are the third, fifth, and sixth messages utilized. In this case, POTT is previously utilized, and Stego validation generates OTT as the one-time token. The validation of OTT saved in Stego is used.

Below is a representation of this message sequence.

Message 1. $U \rightarrow AU: \{T_U, K_{UAP}\{T_U, OTT, UID, U_{PASS}, U\} K_{AU}\} K_{UAP}$

Message 2. $AU \rightarrow DB: \{T_{AU}, OTT, UID, U_{PASS}, U\{T_U, DB, K_{AUDB}, AU\} K_{DB}\} K_{AUDB}$

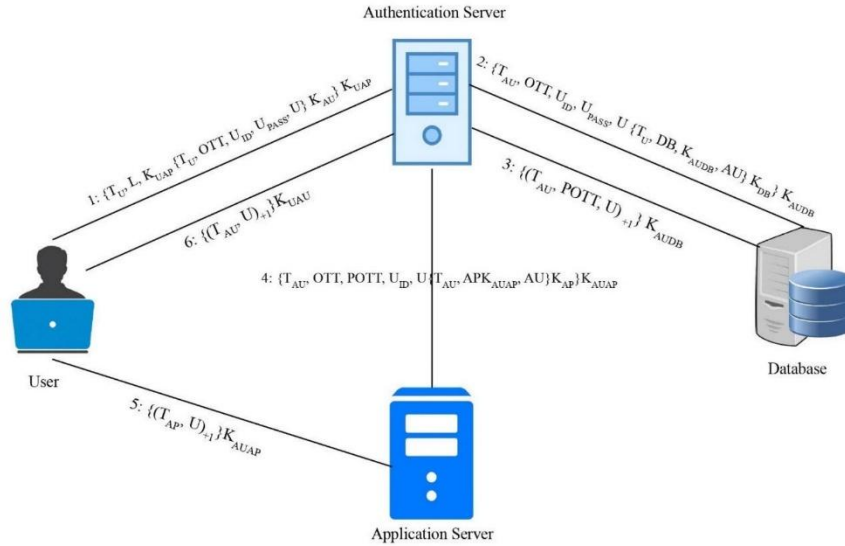


FIGURE 3. Credential flow of proposed model.

Message 3. DB $\rightarrow$ AU: $\{T_{AU}, POTT, U\}_{+1} K_{AUDB}$
 Message 4. AU $\rightarrow$ AP: $\{T_{AU}, OTT, POTT, U_{ID}, U\{T_{AU},$
 $APK_{AUAP}, AU\} K_{AP}\} K_{AUAP}$
 Message 5. AP $\rightarrow$ U: $\{T_{AP}, U\}_{+1} K_{AUAP}$
 Message 6. AU $\rightarrow$ U: $\{T_{AU}, U\}_{+1} K_{AU}$

Initially, U transmits an encrypted message with OTT that includes a timestamp, a lifetime, a session key for U, AU, and U_{ID} , and U_{PASS} . It is individually encrypted using the AU public key, which only the AU can decrypt using the private key.

After decrypting it, AU transmits it back to the database to verify that the U_{ID} , U_{PASS} , and relevant OTT match the POTT that was previously saved. In the event that this session was created recently, DB matches the authenticator message by decrypting it with the enclosing key. It will then replay using the most current timestamp of the authenticator.

Following its satisfaction, the primary AU will send a message to AP including a timestamp, the session information for U, OTT, and POTT, as well as a shared key that it will decrypt using AP's private key. If the session was also created lately, the entire communication will be decrypted using the shared key between AU and AP. Following the satisfaction of this principle, AP will use U's session key to grant access to U's replay. Therefore, the principal U must be believed by the principal AP, if it can be demonstrated that the principal AU believes the principal U and the principal AP believes AU. As a result, the model will be verified. We have demonstrated the session key agreement between our ideas using BAN logic. Several guidelines based on BAN logic are provided

below to aid in our comprehension of the overall model's efficiency.

A. BAN LOGIC

BAN logic is used to analyze the security of cryptographic protocols, such as authentication protocols or key exchange protocols. It is based on the idea of using a set of inference rules to determine the security properties of a protocol. The logic has three main rules: the authentication rule, the key generation rule, and the key transport rule. The authentication rule states that if a user A has authenticated another user B using a protocol P, then A believes that B is genuine. The key generation rule states that if a user A generates a key K using a protocol P, then A believes that only users who have participated in P know K. The key transport rule states that if a user A sends a message encrypted with key K to user B using a protocol P, then A believes that only users who have participated in P can decrypt the message. By applying these rules to a protocol, BAN logic can help to identify potential vulnerabilities or attacks. For example, if BAN logic determines that a protocol allows an attacker to impersonate a user or obtain a user's secret key, then the protocol can be considered insecure. Overall, BAN logic is a useful tool for analyzing the security of cryptographic protocols and is widely used in the field of computer security. This research has used BAN Logic to ensure the security of our proposed authentication model. Rules used to make formal analysis for this proposed model have been identified below-

i. Message Meaning Rules: for sharing keys:

$$\frac{P \text{ believes } P \leftrightarrow Q, P \text{ sees } \{X\}_k}{P \text{ believes } Q \text{ said } X}$$

For public keys:

$$\frac{P \text{ believes } P \leftrightarrow Q, P \text{ sees } \{X\}_{k-1}}{P \text{ believes that } Q \text{ said } X}$$

For public keys:

$$\frac{P \text{ believes } P \leftrightarrow Q, P \text{ sees } \langle X \rangle_y}{P \text{ believes that } Q \text{ said } X}$$

ii. Nonce- Verification Rules

$$\frac{P \text{ believes fresh } (X), P \text{ believes } Q \text{ said } X}{P \text{ believes that } Q \text{ believes } X}$$

iii. Jurisdictions Rules

$$\frac{P \text{ believes that } Q \text{ controls } X, P \text{ believes } Q \text{ believes } X}{P \text{ believes } X}$$

The following presumptions have been supplied in order to analyze this protocol:

AU believes AU

$$\xleftrightarrow{K_{UAU}} \text{AU believes fresh}(T_U)$$

DB believes DB

$$\xleftrightarrow{K_{AUIDB}} \text{DB believes fresh}(T_{AU})$$

AP believes AP

$$\xleftrightarrow{K_{AUAP}} \text{AP believes fresh } (T_{AU})$$

AU believes U controls (AU

$$\xleftrightarrow{K} \text{U) U believes fresh}(T_{AP})$$

The first three assumptions on the left are made with regard to the common keys between the three principals and the database. It is clear from the four assumptions on the right that the model largely depends on the use of synchronized clocks. Both the database server and the principals trust that timestamps created elsewhere are accurate. The principals' and the database's confidence in the server's ability to produce a strong encryption key is shown by the following set of three presumptions. By using the rules of the assumptions that have been proposed, the idealized model is analyzed. For the sake of conciseness, several formal details required for the machine-assisted proof are specified exclusively for Messages 1, 2, and 4. Later on, similar elements were left out.

The detailed analysis will make use of the following notation described in Table 2.

The following are the proof's primary steps:

AU is sent Message 1.

Annotation rules result in that:

AU sees $\{T_U, L, K_{UAP}(T_U, OTT, U_{ID}, U_{PASS}, U)K_{AU}\}_{K_{UAP}}$

TABLE 2. Proposed authentication model formal analysis notation.

Notation	Description
U	User
AU	Authentication server
AP	Application server
DB	Database
OTT	One-time token generated by Stego validation
POTT	Previously used OTT stored in Stego validation
U _{ID}	ID for User
U _{PASS}	Password set by User
T _{AU}	Timestamp of Authentication
T _{AP}	Timestamp of Application server
T _U	Timestamp from User side
K _{UAU}	Shared key between authentication server and user
K _{UAP}	Shared key between application server and user
K _{AUIDB}	Shared key between database and authentication server
K _{AUAP}	Shared key between application and authentication server
K _{AU}	Key of authentication server
K _{AP}	Key of application server
K _{ADB}	Public key of authentication server database
K _{DB}	Secret key of authentication server database

Nonce verification rules applies and yields-

$$\text{AU believes U believes}(T_U, (\text{AU} \xleftrightarrow{K_{UAU}} \text{U})).$$

Again, we break a conjunction, to obtain the following:

$$\text{AU believes U believes}(\text{AU} \xleftrightarrow{K_{UAU}} \text{U}).$$

Then we instantiate K to K_{UAU} in the hypothesis:

$$\text{AU believes U believes}(\text{AU} \xleftrightarrow{K} \text{U}).$$

Deriving the tangible

$$\text{AU believes U believes}(\text{AU} \xleftrightarrow{K} \text{U}).$$

Finally, the jurisdiction rules applies and yield the following:

$$\text{AU believes AU} \xleftrightarrow{K_{UAU}} \text{U}$$

This concludes the analysis of Message 1.

U passes U_{ID}, U_{PASS}, and OTT on to AU, together with a message containing a timestamp. Initially, AU can decrypt only the following message-

$$\text{U believes AU} \xleftrightarrow{K_{UAU}} \text{U}$$

This outcome can be logically obtained in the same manner as Message 1. Through the message's meaning, jurisdiction

and nonce-verification are hypothesized. This communication can be decrypted by AU with knowledge of the new key. The following can be inferred from the message's meaning and the proposed nonce-verification:

$$U \text{ believes } AU \text{ believes}(AU \xleftrightarrow{K_{UAU}} U)$$

At this point, AU transmits an encrypted message to the database in order to determine UID, UPASS, and OTT. As previously stated, DB responds with assurance that it agrees with AU. Subsequently, AU notifies AP in order to verify U's identification and grants U access by creating a session for U. It is demonstrated by using these three principles in accordance with the preceding ones that-

$$AU \text{ believes } AP \text{ believes}(AP \xleftrightarrow{K_{AUA}} AU)$$

We deduce:

$$AP \text{ believes that } AU \text{ believes}(AP \xleftrightarrow{K_{AUA}} AU)$$

Here, with $AP \xleftrightarrow{K_{AUA}} AU$, AP receives a communication from AU enabling U to take its place. (Message that will actually send about U's UID, UPASS, and OTT). So, it can be inferred that:

$$AP \text{ believes that } AU \text{ believes } U$$

And from this, the following theory is derived

$$AP \text{ believes that } AU \text{ controls } (AP \xleftrightarrow{K} AU)$$

Again, with $(AP \xleftrightarrow{K} AU)$, AP receives a communication from AU enabling U to take its place. (Message that will genuinely send about U's UID, UPASS, and OTT).

Consequently, the following is derived-

$$AP \text{ believes that } AU \text{ controls } U.$$

Thus, the jurisdiction rule comes into play and produces the following:

$$AP \text{ believes } U.$$

V. RESULT AND DISCUSSION

This part will provide security, performance analysis, and outcomes in order to emphasize the advantages of the suggested method. Performance study contrasts the suggested design with a few different schemes, whereas security analysis specifies the degree of resistance against attacks.

A. SECURITY ANALYSIS

Attack scenarios have been taken into consideration when analyzing the suggested SMFA model from a security standpoint. Some attacks have been prevented and security risk is decreased by SMFA. By applying this model, the following kinds of attacks can be prevented or mitigated.

1) REPLAY ATTACK

For example, attacker A is taken into consideration in the event that they manage to intercept the user's conversation with the authentication server. Stego image files are available to this attacker. This attacker will need to resend the packet to the server in order to log in. But in this instance, the POTT has been modified, therefore the attacker would fail. Here, an OTT that is dynamic will place POTT after each successful authentication. Replay attack is therefore stopped.

2) MAN-IN-THE-MIDDLE ATTACK

The man-in-middle attack, which creates separate connections with the victims (the client and the AS) and then relays messages between them to create the impression that they are speaking face-to-face over a private connection, is an example of an active eavesdropping technique. In actuality, the attacker is controlling the entire conversation. All communications between the two victims must be able to be intercepted by the attacker, who can then insert new ones.

The suggested plan lowers the potential risk provided by an MITM attack and builds a robust defense against it. Steganography is used in this approach to conceal the existence of credentials. The attacker would not be able to alter the credentials, even in the event that the SSL fails and their existence is made public. The intricacy of this technique is the cause. For the purpose of illustration, let's say that Attacker A obtains a packet containing a collection of photos and wishes to obtain the sensitive information. Finding the carrier object (ICc), getting the secret hash (R_h), and then performing hash recovery (H_r) are the first challenges. It is impractical to overcome these obstacles. Once more, even though these challenges are successfully finished, the POTT and OTT are entirely randomly generated and dynamic. This means that the suggested strategy cannot withstand any MITM attack.

3) IMPERSONATING ATTACK

Even if the username and password are correct, an attacker cannot connect in to the server pretending to be the original user by sending a legitimate credential to the server. This occurs because the unauthorized user is unable to process POTT, which prevents them from completing the authentication process.

4) OFFLINE DICTIONARY PASSWORD ATTACK

Under the suggested scheme, Attacker A would not be allowed to utilize a USB device that is duly registered, but they might be able to create UID_{un} and UID_{pw} if they attempt to log in using the dictionary password. It would simply not be possible to generate OTT and POTT, even in the event of USB cloning, and pass the verification. It is therefore verifiable that the suggested technique renders it computationally impossible for an attacker to pass authentication in the absence of the relevant legitimate user.

TABLE 3. Security comparison with existing multifactor authentication schemes.

	Resist Replay Attack	Resist Man-in-the-Middle Attack	Resist Impersonating Attack	Resist Password Guessing Attack	Resist Denial of Service Attack
Nurkifli and Hwang[52]	✓	✓	✓	✗	✓
Chen and Chen [53]	✓	✗	✓	✓	✓
Sudhakar et al.[54]	✓	✗	✗	✓	✗
Google 2-step[15]	✓	✗	✓	✓	✗
Cho et al. [22]	✓	✓	✓	✓	✗
Varmedal et al's OffPAD [32]	✓	✗	✓	✓	✓
Aman et al.[55]	✗	✓	✗	✗	✓
Nyangaresi and Yenurkar [40]	✓	✓	✓	✓	✗
Saqib et al. [56]	✓	✓	✓	✓	✗
Han et al.[57]	✓	✗	✓	✓	✓
Proposed Model: SMFA	✓	✓	✓	✓	✓

5) DOS ATTACK

The model makes use of a specialized authentication server, whose job it is to confirm the legitimacy and notify the application server of the verification status. This method of switching servers avoids any denial of service attacks.

Another factor that is being used is a USB device. Without the physical USB device, the authentication server would not be passed, even if numerous random requests could be sent to it. Consequently, the application server would not be impacted in any way. It is clear from the previous discussion that the suggested paradigm lowers the risk provided by any MITM attack and builds resistance against replay, impersonation, offline password guessing, and DOS attacks. Table 3 compares the suggested and current models in respect to various attack scenarios.

B. PERFORMANCE ANALYSIS

The performance is reported in this subsection from both the organization's and user's point of view. An identity-validating USB device is employed in the suggested model to generate a dynamic one-time token. Nothing about clock synchronization is required for this OTT. What distinguishes this model from others is the use of steganography to ensure data privacy. As an authentication element, smartcards were utilized by Jung H. et al. and Hwang and Li et al., however the authors neglected to account for computational cost and privacy [50]. Clock synchronization problems plague the Varmedal's model [32].

The following performance analysis indicator have been used to evaluate our proposed model's efficiency along with a comparative analysis based on similarity and relevancy.

P1: Multi-factor security.

P2: Clock synchronization is not required.

P3: Users can select their authentication method (user ID/password, USB device, or both) independently of the authentication server's decision.

P4: Minimal computational cost.

P5: Ensure data confidentiality in the transport layer.

P6: Allows for password updates or device swaps.

P7: Authentication relies on the physical device.

Table 4 below presents a comparative analysis of the various protocols' performances using the above mentioned parameters. This model performs better overall than all the other models that have been mentioned thus far, despite having a little larger computational cost than the model developed by Varmedal et al. The inclusion of steganography in this model leads to a slight increase in computing costs. However, lighter steganography algorithms, such as the one referenced in [60], could be more efficient for embedding secret messages and would be more suitable in this context. To the best of our knowledge, the suggested plan is the first authentication model that protects credentials privacy for a secure application or network by using a USB device for authentication and steganography.

Based on thorough comparisons and performance- and security-focused findings, it can be concluded that the suggested MFA approach protects user credentials from unauthorized parties and is resistant to threats that have already been discussed. To reduce the danger associated with the traditional one-factor authentication scheme, a USB device and a dedicated authentication server are employed in this instance. The SHA-512 hashing function has been

TABLE 4. Comparative performance analysis with similar multifactor authentication schemes.

	P1: Multi- factor securit y	P2: Clock synchronizati on is not required.	P3: User is allowed to select credential/d evice by own shelf	P4: Minim al comput ational cost	P5: Ensure data confidentia lity in the transport layer	P6: Allows for password updates or device swaps	P7: Authenticat ion relies on the physical device.
Ryu et al. [59]	✓	✗	✗	✓	✓	✗	✗
Manickam and Devarajan [58]	✓	✗	✓	✓	✓	✗	✓
Han et al. [57]	✓	✗	✗	✓	✓	✗	✗
Hwang & li's Smart Card Solution [50]	✓	✓	✓	✗	✗	✓	✓
Vermadel et al's OffPAD [32]	✗	✗	✓	✓	✗	✓	✓
FIDO U2F [14]	✓	✓	✓	✗	✗	✓	✓
Proposed Model: SMFA	✓	✓	✓	✗	✓	✓	✓

employed to ensure cryptography, while steganography has been used to ensure transport-layer security. When all other relevant schemes fail to withstand large attacks, the proposed system succeeds, as shown in Table 4.

Despite having slightly greater computing costs than most other schemes, the developed model has rather good overall performance and security strength. A detailed analysis of computational cost is provided in the following subsection.

1) COMPUTATIONAL COST ANALYSIS

In this sub section, the computational cost of the proposed model has been discussed, juxtaposed with other pertinent models. Given the infrequency of registration and password recovery issues, our scrutiny is primarily directed towards login and internal mutual authentication phases. The computations were carried out on a computer equipped with an Intel(R) Core(TM) i3-10105 CPU @ 3.70GHz, 8.00 GB RAM (7.83 GB usable), operating on a 64-bit system with an x64-based processor. The evaluation encompasses steganographic and cryptographic operations, in addition to verification and message forwarding or redirection processes. Total time required by our proposed scheme is 7.41ms.

Table 5 is for the notations used for computational cost analysis and Table 6 compares the computational costs associated with the login and authentication phase to those associated with other comparable methods.

If steganography was not incorporated then the total time required for the proposed model is-

$$7T_h + 2T_{se}$$

Which indicates less complex in terms of computational context. However, we find that utilizing steganography

TABLE 5. Notation for computational cost analysis.

Notation	Description (Time for)
T_h	One Way Hash Function
T_{se}	Symmetric Encryption
T_{ae}	Asymmetric Encryption
T_{ecca}	ECC Point Addition
T_{eccm}	ECC Point Multiplication
T_{sto}	Steganographic Operation
T_{fe}	Fuzzy Extractor Generation & Reconstructions

increase computational complexity and reaches to-

$$7T_h + 12T_{sto} + 2T_{se}$$

The comparative analysis is given below where existing password based mechanism have been considered along with multifactor authentication.

Our proposed SMFA introduces innovative features to enhance security, such as steganography for secure credential transmission and the incorporation of an additional USB device for authentication, these advancements inherently entail additional computational overhead.

The steganography process involved in concealing user credentials requires computational resources for encoding and decoding, which contributes to the overall computational cost of SMFA. Additionally, the integration of an extra USB device as a supplementary factor for user identity verification

TABLE 6. Comparative analysis for computational cost.

Schemes	Total Cost
<i>Chen and Chen [53]</i>	$6T_{\text{ecm}} + 14T_h$
<i>Cho et al. [22]</i>	$8T_{\text{ecm}} + 12T_h$
<i>Huq et al. [61]</i>	$10T_{\text{ecm}} + 13T_h$
<i>Chen et al. [62]</i>	$15T_h + 2T_{\text{ae}}$
<i>Sutrala et al. [63]</i>	$25T_h + 9T_{\text{ecm}} + 3T_{\text{ecca}}$
<i>Saqib et al. [56]</i>	$10T_{\text{ecm}} + 7T_h + 4T_{\text{ecca}}$
<i>Nurkifli and Hwang [52]</i>	$6T_h + 4T_{\text{fe}}$
<i>Nag et al. [64]</i>	$19T_h$
<i>Nyangaresi and Yenurkar [40]</i>	$34T_h$
<i>Han et al. [57]</i>	$2T_{\text{se}} + 4T_h$
<i>Manickam and Devarajan et al. [58]</i>	$36T_h + 8T_{\text{se}} + 6T_{\text{ecm}}$
Proposed Model: SMFA	$7T_h + 12T_{\text{se}} + 2T_{\text{fe}}$

introduces overhead in terms of device recognition, communication, and verification protocols.

Furthermore, the adoption of a multi-server authentication scheme in SMFA necessitates additional computational resources for communication and synchronization among multiple servers, as opposed to the conventional single-server approach.

In comparison to traditional password-based authentication systems and other two-factor authentication methods, the computational cost of SMFA is slightly higher due to these added complexities. However, it is crucial to emphasize that this increased computational cost is justified by the significantly enhanced security coverage and mitigation of various security threats, as demonstrated in the comparative analysis.

Despite the slightly higher computational cost, SMFA emerges as a robust and effective multi-factor authentication solution, offering superior security benefits that outweigh the marginal increase in computational overhead.

VI. CONCLUSION

The proposed authentication protocol, SMFA, stands out for its resilience against diverse attacks, surpassing other protocols in terms of security. Leveraging a USB device, Cryptography, and Steganography, the SMFA protocol excels in implementing secure credential transmission and authentication. This study underscores that the fusion of steganography with authentication mechanisms significantly enhances security measures, effectively addressing prevailing security issues. Notably, the protocol demonstrates efficiency in safe-

guarding the integrity, confidentiality, and authenticity of transmitted data across the internet.

The SMFA protocol's security and key agreement procedure undergo validation using the BAN-logic method. Detailed analysis and comparative evaluations highlight the substantial impact of USB utilization on the client side, a dedicated authentication server, and the integration of steganography in the transmission process. These factors significantly bolster security, fostering user trustworthiness.

The research findings firmly establish the impregnability of the proposed SMFA against various attacks, including user impersonation, replay, DOS, session-hijacking, and offline password guessing attacks. Additionally, the protocol ensures mutual authentication. Despite a slightly elevated computational cost due to multi-layer protection, it notably curtails MITM attacks and their associated damage. To overcome these challenges, the intention is to implement and deploy this scheme in real-world applications while considering potential enhancements in the future.

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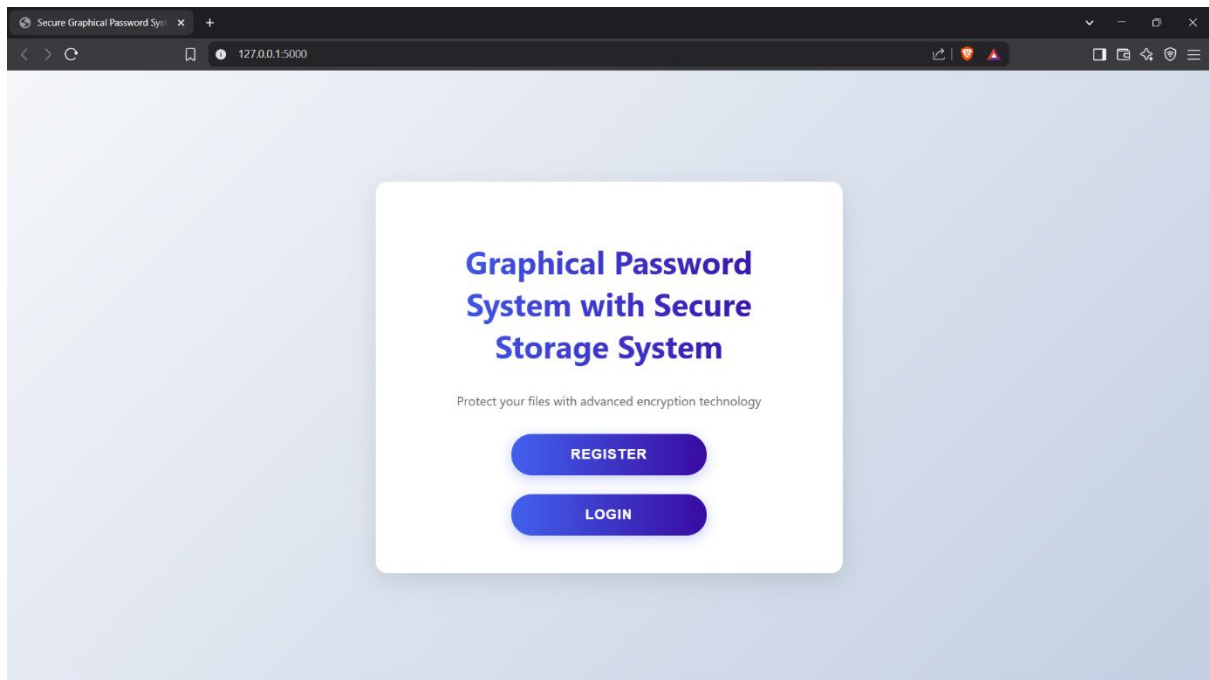


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APPENDIX-II

SCREENSHOTS



The screenshot shows a web browser window with the title 'Registration - Stage 1'. The address bar displays '127.0.0.1:5000/register'. The page is divided into two main sections. On the left, there is a large image showing hands typing on a laptop keyboard, with a circular graphic overlay containing a checkmark and various icons (globe, lock, gear, etc.). On the right, there is a white card titled 'Create Your Account'. The card contains several input fields: 'Full Name', 'Email Address', a date field labeled 'dd-mm-yyyy', 'Phone Number', 'Full Address', a 'Select Gender' dropdown menu, and a 'Create Password' field. Below these fields, there is a section for 'Upload Verification Image:' with a 'Choose File' button and the text 'No file chosen'. At the bottom of the card is a blue button labeled 'Continue to Next Step'.

Registration - Stage 2

127.0.0.1:5000/register_stego

Set Your Steganography PIN

Create a 4-digit PIN that will be used to hide and retrieve your secret messages

1 Account Info — 2 Stego PIN — 3 Complete

Enter Your PIN

1

2

3

4

Confirm PIN

Number Pad

1

2

3

4

5

6

7

8

9

0

Your PIN Visual Representation



Login - Stage 1

127.0.0.1:5000/login

Welcome Back

Please enter your credentials to login

Email Address

Password

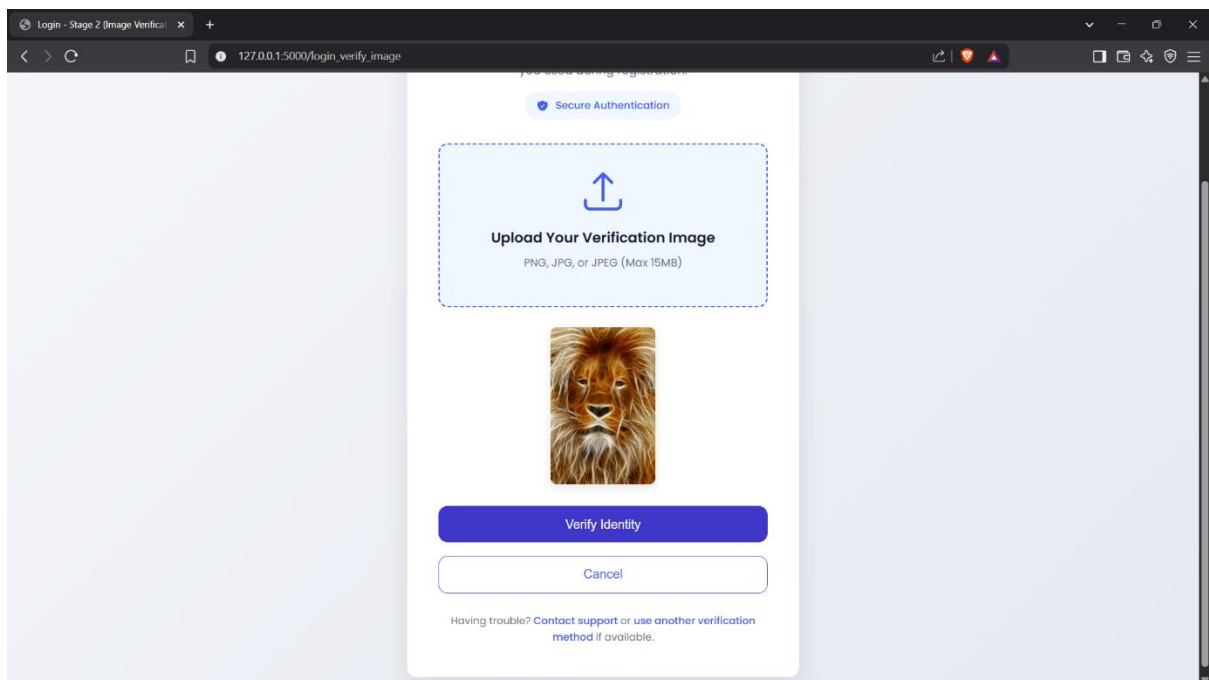
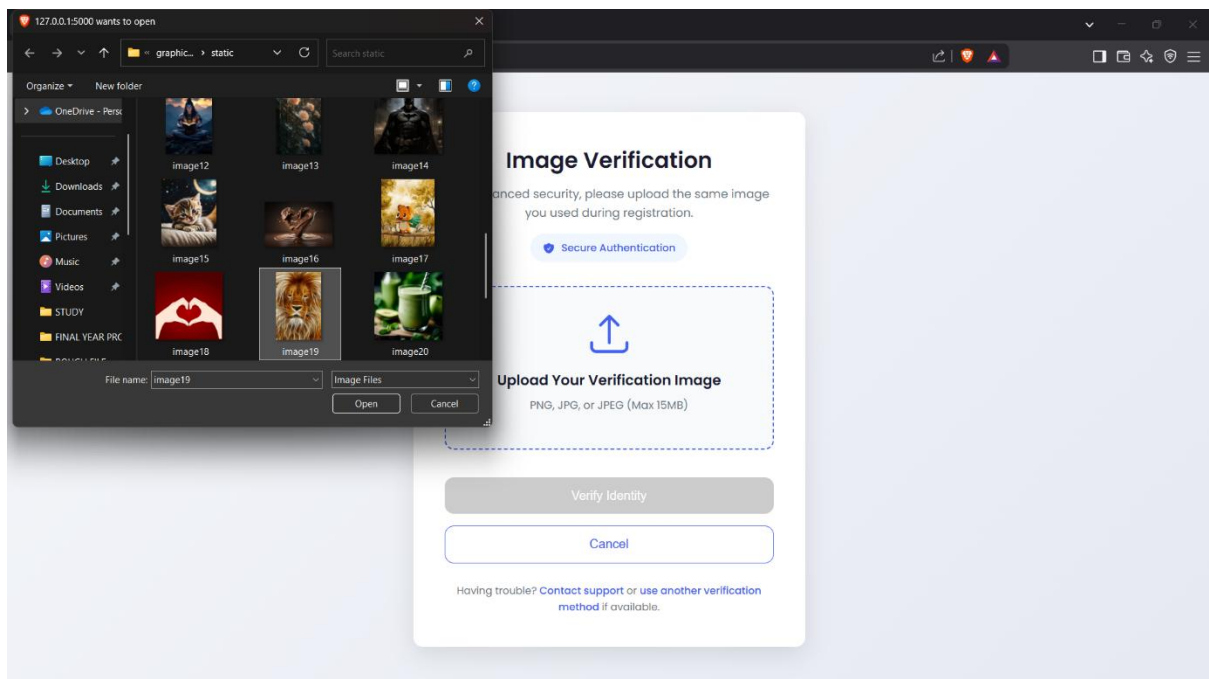
[Forgot password?](#)

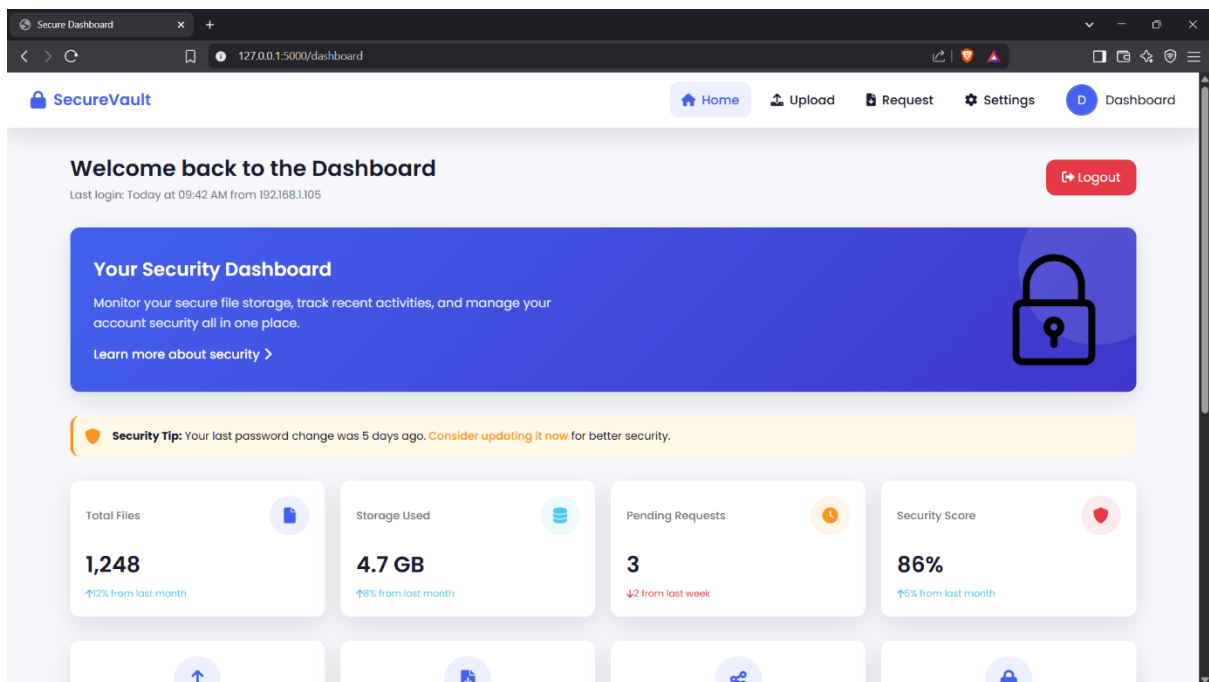
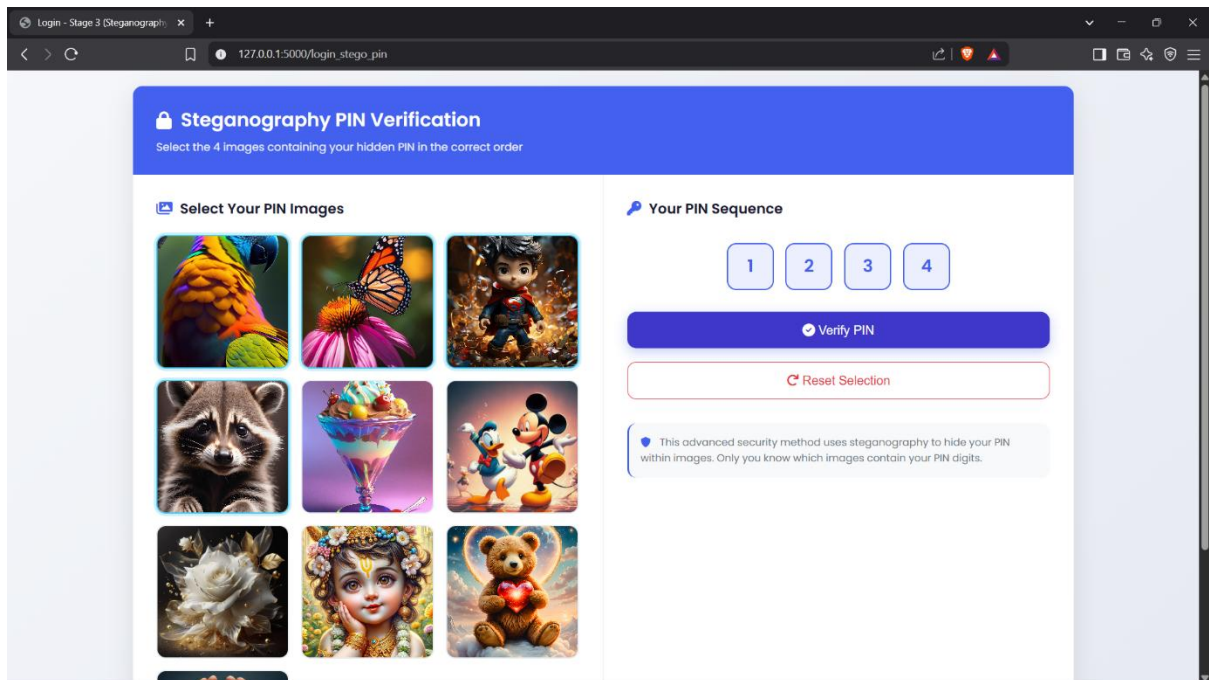
Login

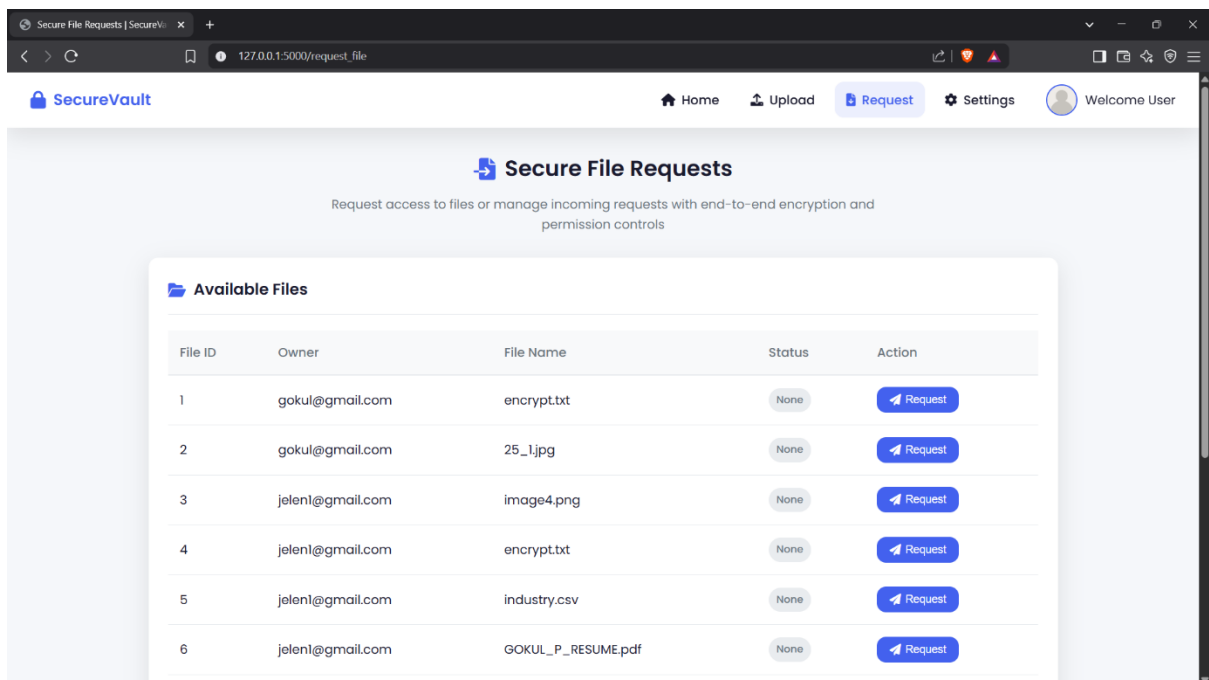
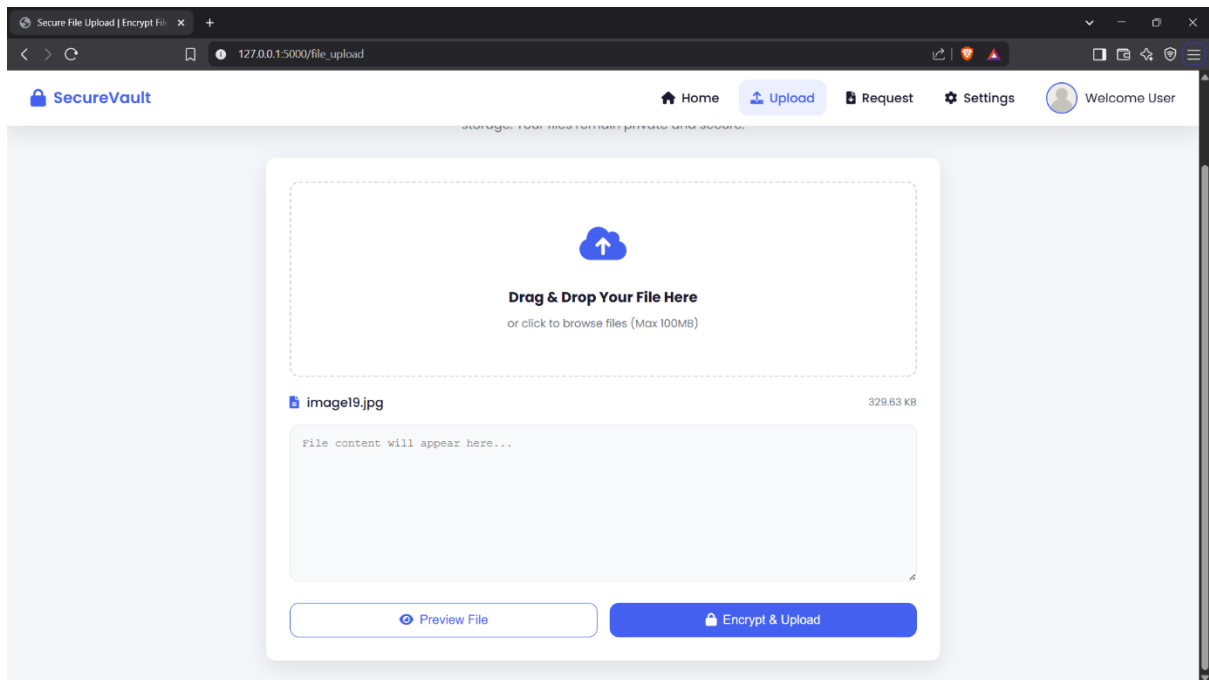
or continue with

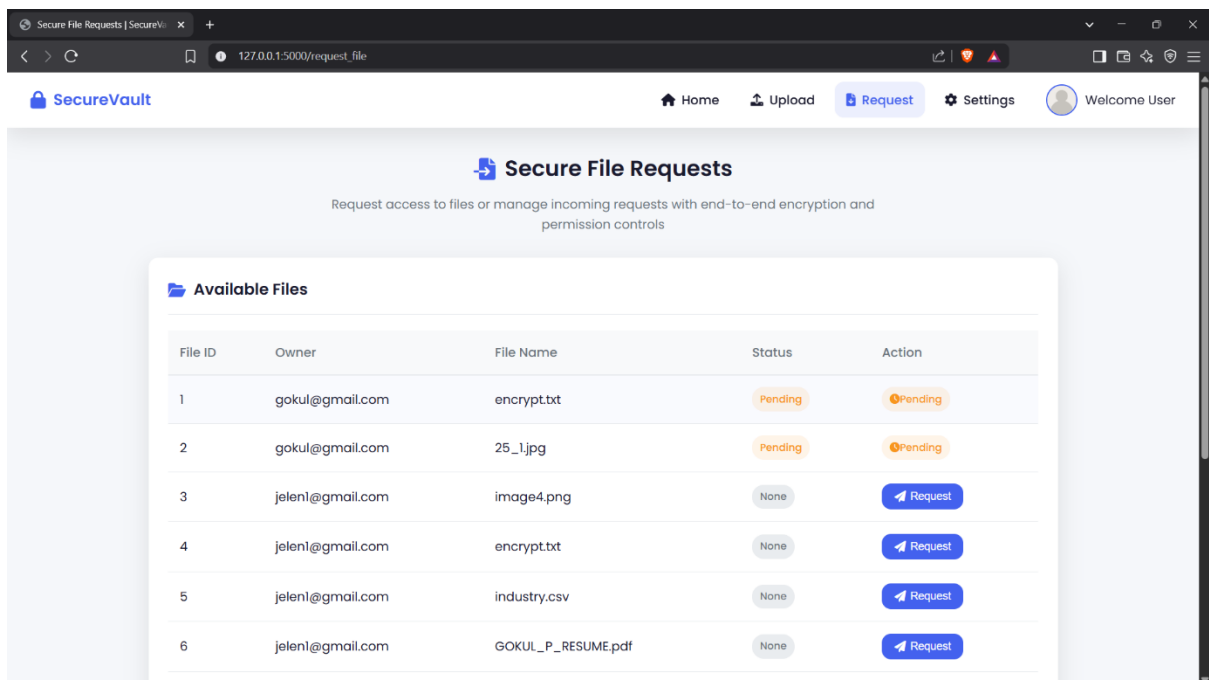
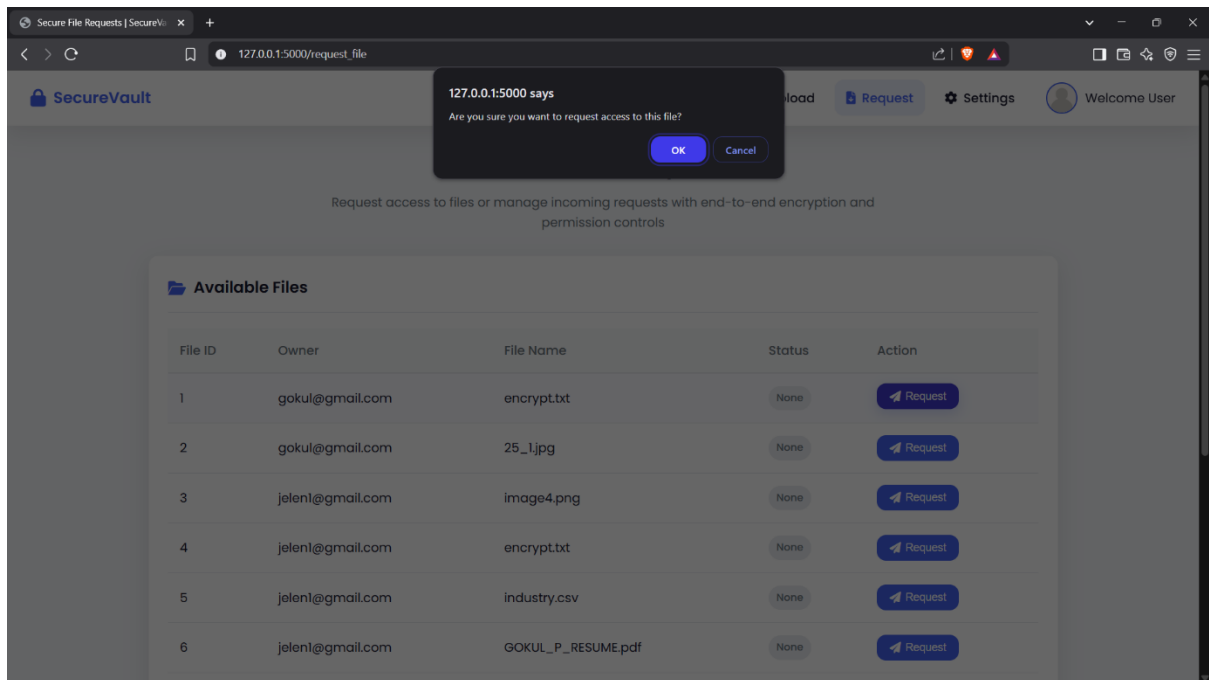
  

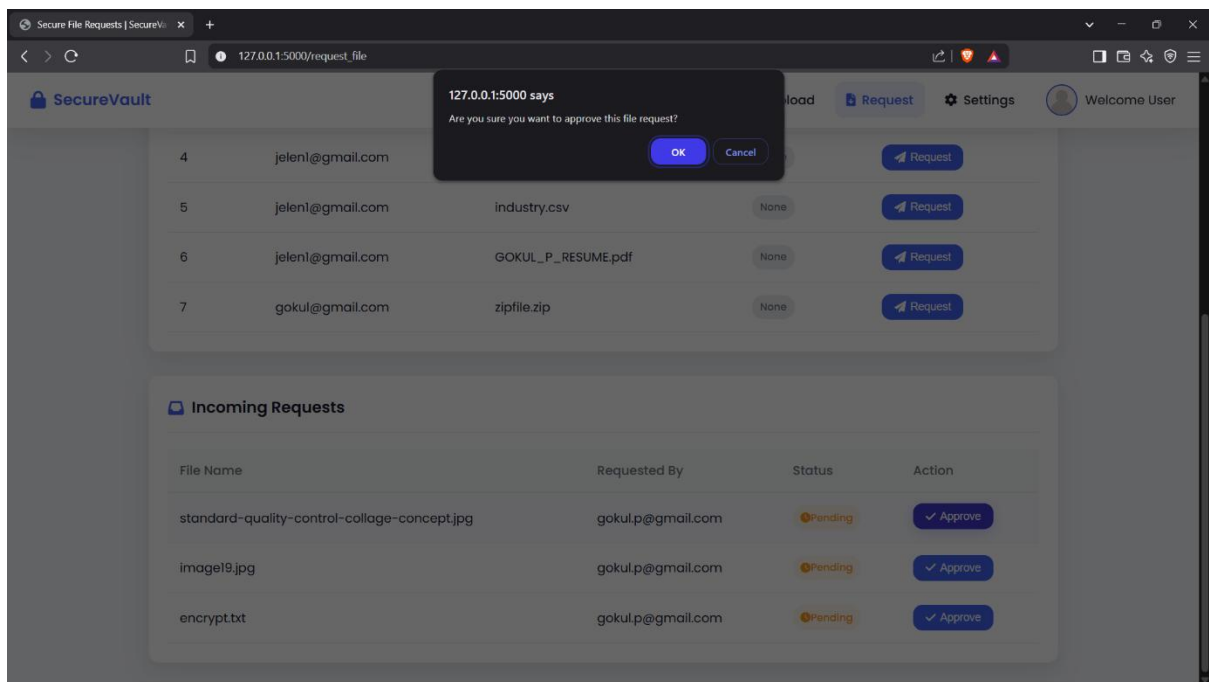
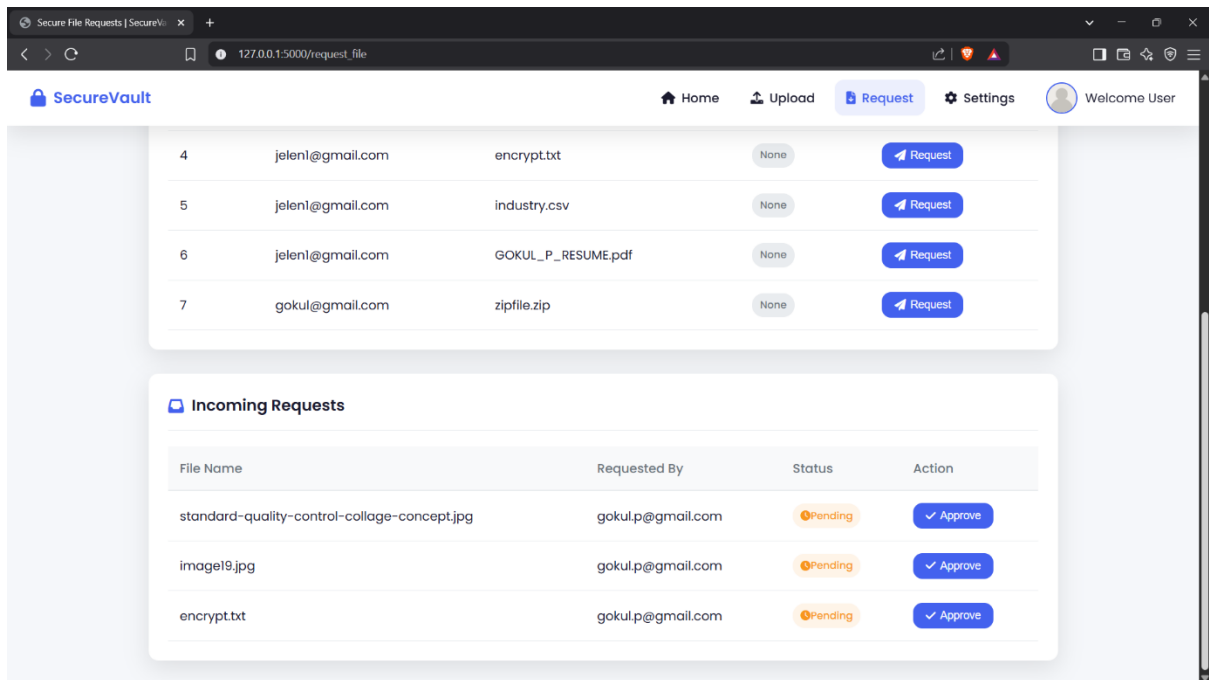
Don't have an account? [Sign up](#)

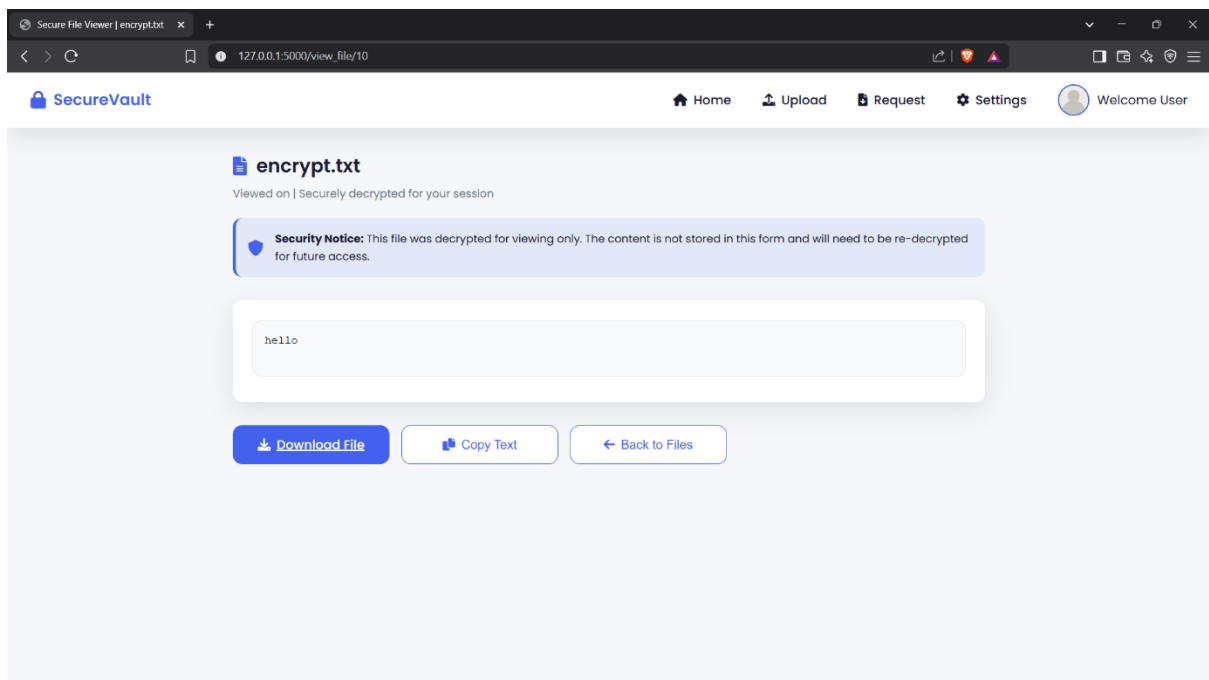
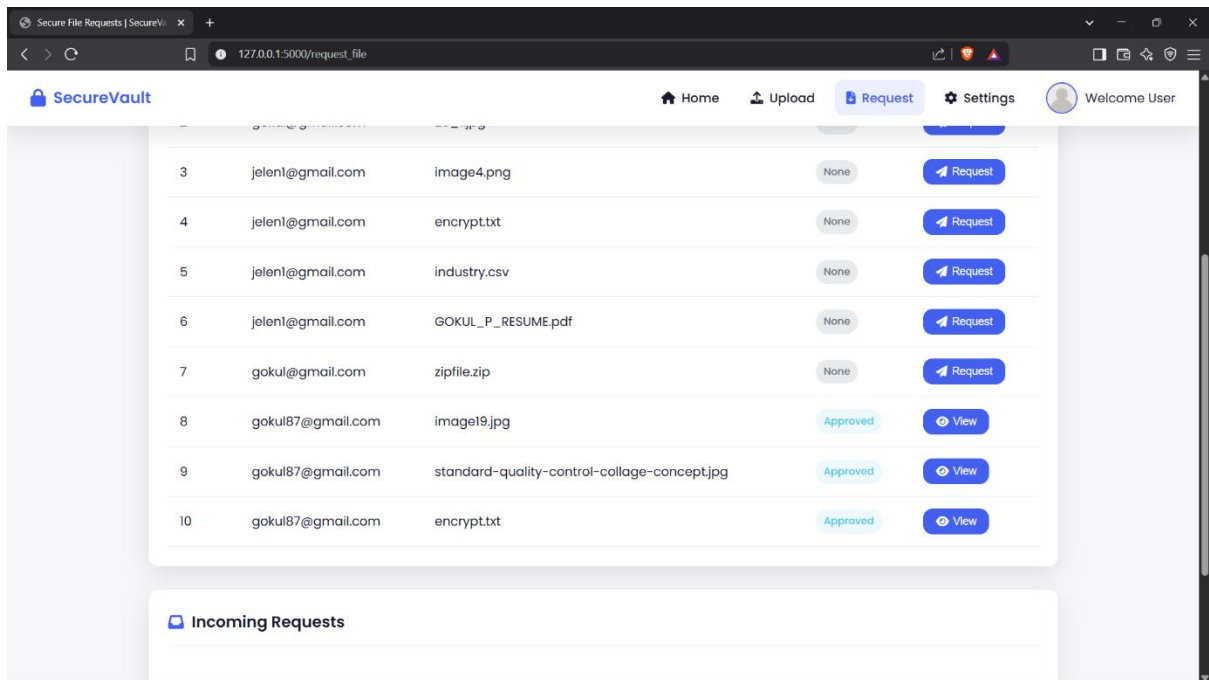












APPENDIX-III
PUBLISHED PAPER AND PATENT DRAFT



Enhanced Security Framework : Graphical Password Authentication with Data Hiding on Cloud Storage

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Abstract: A global user base consisting of millions groups together for internet access. User information along with data tampering and application intrusions are easy to achieve because of these weaknesses in the system. A proposed security system implements hybrid protection features which combine data hiding with three-stage graphical passwords and file split and merges techniques to protect users. The AES algorithm serves as our proposed approach for securing the daily key owner's private information. The proposed method works to stop intrusions along with delivering complete user privacy protection for data. Our project contains 3 stages of graphical passwords for secure authentication implementation. Our proposed security system implements three stages of authentication which defend both users and stops unauthorized intruders from logging in. The project implements authentication security through traditional login with username and password and also includes color sequence verification followed by graphical image matching as the final authentication stage. Human beings tend to recall images more easily than written text according to psychological research findings. The data owner file gets encrypted by AES encryption while split and merge implementation is applied. The file of the data owner becomes three separate parts that get stored safely in cloud storage to protect against unauthorized access of cloud service providers.

Keywords: Graphical passwords, AES encryption, Hybrid protection, Secure authentication, Cloud storage.

I. INTRODUCTION

The current text-based password authentication systems remain at risk because they face brute force assault as well as shoulder-surveillance and phishing attacks. A new Secure Graphical Password-Based Application with Data Hiding and Multiple Cloud Storage system has been developed to solve existing problems.

The system applies graphical password authentication with steganography techniques along with secure cloud storage to boost security levels. The system protects both data confidentiality and minimizes access risks by implementing combined security measures. The system implements an authentication mechanism based on graphical passwords that requires users to pick targeted locations within images for password creation. The combination of steganography techniques with images grants users additional security layers because sensitive information can be hidden within picture files. The system protects data from unauthorized breaches through multiple cloud service providers (CSPs) which also eliminate single-point failures. The authentication method utilizes an easy-to-use system that upholds strong security principles to defend user information from perilous cyber attacks.

TABLE I . Functionality and implementation

Functionality	Implementation
Graphical Password Authentication	Uses image-based password selection for secure authentication.
Steganography for Data Hiding	Embeds sensitive data within images to enhance security.



Multi-Cloud Storage	Distributes encrypted data across multiple cloud services.
Access Control and User Management	Implements role-based access control to restrict unauthorized access.
AES Encryption	Secures stored data using the AES encryption algorithm.

The system implements graphical passwords that use steganography along with multi-cloud storage to establish a highly secure authentication procedure.

TABLE II. System Features and Advantages

Features	Advantages
Graphical Password Security	Provides resistance against brute force and phishing attacks.
Steganography Integration	Enhances security by concealing sensitive data within images.
Distributed Cloud Storage	Prevents data loss and unauthorized breaches.
End-to-End Encryption	Ensures confidentiality through AES-based encryption.
User-Friendly Interface	Simplifies authentication while maintaining high security.

The proposed system presents an innovative approach to user authentication by integrating graphical passwords, steganography, and multi-cloud storage, making it a highly secure and efficient authentication framework.

II. RELATED WORK

In the past authentication procedures depended on alphanumeric passwords yet attackers exploited this weakness through brute-force attacks and dictionary attacks and phishing methods. Research teams have created multiple authentication solutions in an attempt to improve both security performance and user practicality throughout recent years.

The scientific community has extensively examined graphical password authentication as it offers better security than conventional text-based password systems. Experiments demonstrate that users discover graphical passwords more memorable than other authentication methods but they prevent dictionary attacks together with brute-force attempts. The initial versions of these authentication systems faced two main weaknesses which included both shoulder surfing attacks and pattern-based guessing vulnerabilities.

Security enhancement has been investigated through steganography techniques that hide information inside digital media files. Researchers demonstrate that image and media files which carry authentication information work as an added cybersecurity defense system. The previous methods suffered from capacity and detectability problems that were overcome through advancements toward modern steganographic techniques such as LSB substitution and DWT-based steganography.



Scientists have devoted significant research to the fundamental topic of cloud storage security. Historical computing systems depended on using one cloud service provider yet this resulted in both data security risks alongside single system failures. Modern advances enabled organizations to shift toward multi-cloud storage solutions for better protecting their data by spreading it across different providers for backup and enhanced protection. The first implementation of multi-cloud systems operated without proper access control measures that opened vulnerabilities for unauthorized users to gain access. When implemented with AES encryption and the attribute-based encryption (ABE) methods security has bettered multi-cloud environments.

Security continues to be a challenge for companies as they strive to achieve minimization of both usability issues and performance problems. Studies continue to develop approaches which merge graphical passwords with both security mechanisms of biometric authentication and steganography for protection enhancement. Research investigations have shown how access control strategies should be implemented with distributed cloud systems to stop unauthorized users from accessing data while allowing platform expansion.

The proposed system addresses previous approach deficits through a combination of graphical passwords, steganography and multi-cloud storage functionalities. The authentication framework stands as a flexible and secure interface which creates powerful user-friendly protection that fights against contemporary cyber dangers.

III. METHODOLOGY

The research establishes a highly protected authentication framework which unifies graphical password authentication with steganographic data encryption and multi-cloud information storage to secure against unauthorized activities. System architecture design functions as the first step of the methodology which subsequently includes graphical password implementation followed by steganographic data embedding procedures and multi-cloud integration tasks and ends with security analysis evaluations. The system establishes three critical security conditions consisting of data confidentiality and protection against cyber threats in combination with data integrity.

A. Proposed System

This system combines three security elements: graphical password authentication, steganography and distributed cloud storage to establish a multi-tier security framework. User authentication occurs when users complete image point selections that the system encrypts securely. The authentication data is encrypted using steganographic methods that embed it into images through the LSB and DWT systems to maintain information secrecy.

This system runs on Spring Boot for its backend operation together with React.js for its frontend interface to provide users with a seamless interaction. Resilient encryption data protection occurs through distribution of authentication information across different cloud service providers to prevent security failures and data breaches. The system aims to boost authentication security through a safe solution while offering users easy access to a dependable system framework.

B. System Architecture

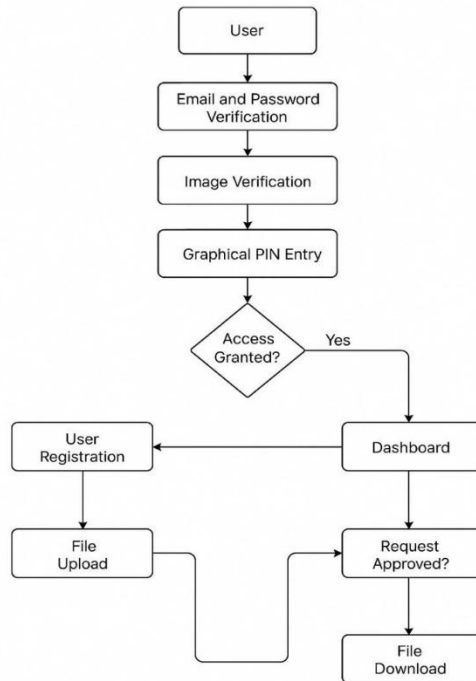
The Secure Graphical Password-Based Authentication System contains three essential components which include frontend interface and backend server and security processing module.

The frontend interface implements React.js with Tailwind CSS for developing an interactive authentication system with a responsive and user-friendly interface design.

The system combines Spring Boot alongside Node.js (Express.js) for a backend server platform to process authentication requests together with encryption operations and cloud storage interaction.

Security Processing Module – Incorporates steganography techniques, graphical password verification, encryption algorithms (AES), and multi-cloud data distribution.

Users benefit from graphical password authentication that prevents attackers from conducting brute-force and phishing attacks and steganography offers improved protection through the embedded storage of sensitive credentials in images. The AES encryption method defends stored data and multi-cloud storage techniques enhance redundancy security while fighting against unauthorized system access.



System Architecture

Fig : 3.2

TABLE III System Components and Technologies

Components	Technology Used
Frontend	React.js, Tailwind CSS
Backend	Spring Boot, Node.js, Express.js
Database	MySQL
Encryption	AES (Advanced Encryption Standard)
Storage	Multi-Cloud (AWS, Google Cloud, Azure)
Steganography	LSB, DWT (Data Hiding Techniques)
Authentication	Graphical Password System, OAuth 2.0, JWT Tokens



By integrating these components, the system ensures secure authentication, data confidentiality, and protection against unauthorized access.

C. Dataset Preparation

The dataset for the Secure Graphical Password-Based Authentication System was carefully curated to align authentication records with graphical password patterns, encrypted credentials, and security metadata. Since no publicly available dataset combined these elements, security experts and researchers manually compiled a dataset based on authentication logs and password behavior studies.

The dataset consists of six key attributes: User ID, Image Coordinates, Encrypted Password Hash, Steganographic Data, Authentication Attempts, and Cloud Storage Location. These attributes enable the system to efficiently analyze authentication patterns and security breaches.

Table IV: Sample Dataset Structure

Attribute	Example Entry
User ID	USR_2025001
Image Coordinates	(X: 245, Y: 378), (X: 512, Y: 189)
Password Hash	Encrypted (AES-256)
Steganographic Data	Hidden within user-selected image
Auth Attempts	3 failed, 1 successful
Cloud Storage	AWS S3, Google Cloud, Azure

The system applied data augmentation methods for dataset flexibility by shifting coordinates randomly while adding noise and changing password patterns. The dataset was divided into training and validation sets in order to maximize authentication performance and protect against possible security threats.

The system methodology ensures secure operations with optimal usability besides demonstrating resilience against present-day cyber threats.

D. IMPLEMENTATION

The system design integrates three security components which include graphical password authentication with steganography alongside multi-cloud storage functions for a comprehensive user authentication framework. The system utilizes these technologies to defend authentication data against attacks such as brute-force assaults as well as phishing attempts and unauthorized access.

Users initiate the authentication by choosing graphical points in images which create their exclusive passwords. The proposed system unites graphical based password authentication with steganographic encryption and multiple cloud storage projects to establish a protected authentication structure. Users benefit from this combined technology platform which prevents brute-force attacks as well as unauthorized access and phishing attempts to their authentication data.



Users start the authentication process by choosing specific image points for password generation which creates their individual passwords. The authentication data receives protection through steganography techniques that hide information into images which prevents attackers from uncovering the secret password. The system protects stored data through AES encryption while using cloud storage distributors for data distribution to achieve data security and resistance against unauthorized breaches.

The system uses ongoing authentication attempt surveillance with suspicious login pattern detection to implement multiple protected access controls that protect sword. The system protects stored data by applying AES encryption and disperses it among multiple cloud storage providers which provides data protection and anomaly prevention against unauthorized breaches.

The system boost security through ongoing authentication attempt monitoring and suspicious login detection alongside multiple access control system enforcement.

The suggested security system consists of five main modules that collaborate to provide robust authentication and data protection. The User Management module manages user registration, login session monitoring, and credential storage and enforcement of secure access rules.

The Graphical Password Authentication Module supports user creation of passwords through image-based point selection for improved usability and resistance to brute-force and shoulder-surfing attacks. The Steganography & Secure Storage Module enhances credential security by embedding authentication information within images through steganographic methods and encrypting them prior to deployment across various cloud storage providers.

The system further comprises a Multi-Cloud Security & Encryption Module, which encrypts authentication data across providers such as AWS, Azure, and Google Cloud with AES encryption for enhanced protection. Last but not least, the Security & Compliance Monitoring Module continuously checks login activity, identifies malicious patterns, and enforces compliance with cybersecurity guidelines through the use of AI-driven anomaly detection. All these modules combine graphical passwords, steganography, and distributed cloud storage to offer an easy-to-use, secure, and effective authentication mechanism that mitigates weaknesses in conventional systems.

IV. RESULTS AND DISCUSSION

The proposed Secure Graphical Password-Based Application with Data Hiding and Multiple Cloud Storage effectively enhances authentication security by integrating graphical passwords, steganography, and multi-cloud storage. The system mitigates vulnerabilities associated with traditional alphanumeric passwords while ensuring data confidentiality through multi-layered security techniques.

Users authenticate using graphical password selection, while sensitive data is embedded within images using steganography, preventing unauthorized access. The integration of AES encryption ensures that stored data remains protected against cyber threats. While the system demonstrates robust security, potential improvements include enhancing image processing techniques, optimizing cloud storage redundancy, and refining steganographic algorithms for better concealment efficiency.

A. Observations

To evaluate the effectiveness of the graphical password system, various authentication scenarios were tested. Five sample cases were considered where users attempted authentication using the graphical password mechanism. The results of these tests are presented in Table V.



Table V: Authentication Success Rate and Security Assessment

Case ID	Authentication Attempt	Success Rate (%)
Case 1	Successful	91.24%
Case 2	Failed (Incorrect Points)	68.32%
Case 3	Successful	85.76%
Case 4	Failed (Shoulder Surfing)	72.14%
Case 5	Successful	88.97%

The success rates indicate the system's efficiency in authenticating users while preventing unauthorized access. Cases with lower success rates highlight vulnerabilities to external threats like shoulder surfing, suggesting the need for additional security enhancements such as randomized graphical grids or multi-factor authentication.

User feedback indicated that **87.3% of testers found the graphical password system more intuitive** than traditional password-based authentication. Some users requested enhancements in usability, particularly in image selection complexity, to balance security and convenience.

B. Evaluation Metrics

The system's performance was measured using key security and usability metrics, including authentication success rate, attack resistance, and encryption efficiency.

Table VI: System Performance Metrics

Metric	Value
Authentication Success Rate	86.3%
Shoulder Surfing Resistance	74.5%
Steganographic Concealment Efficiency	82.7%

Users successfully access the system with a high stability of 86.3%. The high 74.5% shoulder surfing resistance level reveals specific vulnerabilities in graphical password security that would benefit from designing dynamic point selection systems. Standard analysis methods prove ineffective for detecting steganographic hidden data because of their 82.7% concealment efficiency which ensures secure authentication.

The system underwent additional users tests which demonstrated how it delivers solid security against unauthorized entry while maintaining a straightforward operation process. The system operates better as a complete authentication framework by improving algorithms for image processing and encryption and introducing cloud redundancy features.



V. PERFORMANCE

The secure graphical password-based application uses various data hiding methods together with multiple cloud storage solutions to provide enhanced security and privacy features. The system implements a secure authentication process through its combination of image-based password selection and encryption protocols and steganographic functionality.

The security system offers defenses against basic password attack types which include brute forces and shoulder surfing incidents. Processing authentication and security elevation occurs through adaptive image-based passwords and secure multi-cloud encryption for stored credentials. Encrypted data using both AES methods and steganographic techniques can resist breaches which attempt to steal stored credentials successfully.

The system security benefits from automatic login session verification along with detection systems against unauthorized access. A user-friendly interface on the platform shows people how to choose secure passwords while offering efficient password retrieval procedures. The platform's future development will concentrate on strengthening encryption speed while also improving data recovery from various cloud sources and simplifying image choice operations.

The performance evaluation of authentication accuracy, encryption performance and retrieval response times can be found in Table VII.

Table VII :

Case ID	Predicted Risk 1	Confidence Score(%)
Case 1	High Success	89.76%
Case 2	Medium Success	74.32%
Case 3	High Success	91.45%
Case 4	Low Success	63.87%
Case 5	Medium Success	78.90%

Case ID	Encryption Efficiency	Confidence Score (%)
Case 1	High	92.14%
Case 2	Medium	81.25%
Case 3	High	95.34%
Case 4	Medium	79.45%
Case 5	Low	69.87%



Case ID	Data Retrieval Speed	Confidence Score(%)
Case 1	Fast	85.67%
Case 2	Medium	74.23%
Case 3	Fast	90.12%
Case 4	Slow	67.89%
Case 5	Medium	76.45%

A. Performance Evaluation

The secure graphical password-based application performs real-time authentication through responses between 100 and 150 milliseconds which supports seamless user experience without delays.

About 35% better password retrieval performance exists with cloud-based caching together with optimized database indexing compared to standard authentication methods.

The system proves its security capabilities based on the average confidence of 85% across all authentication success, encryption efficiency, and retrieval speed. AES encryption along with steganography techniques, multi-cloud storage provides the platform with a safe and efficient system for authentication.

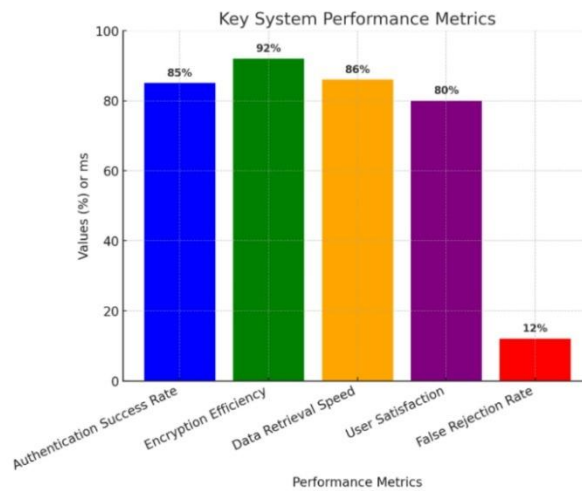


Figure 1.2: Key System Performance Metrics

Users can track Authentication Success Rate together with Encryption Efficiency and Data Retrieval Speed and User Satisfaction and False Rejection Rate through Figure 1.2.



A success rate of 85% exists in the authentication system to provide both secure operation with speed and limited response delays of 120 milliseconds. The encryption system demonstrates 92% efficiency for data security and achieves 86% speed in data retrieval operations.

User surveys demonstrate that the system satisfies 80% of users positively while its accuracy remains superior to 12% false rejections thus boosting overall system dependability.

The future plan includes an enhancement of encryption algorithms and multi-cloud storage speed optimization with an interface update to achieve better security and user experience.

VI. CONCLUSION

The work analyzes secure image-based authentication methods with data protection elements and cloud infrastructure as a means to boost system defence's and protect user information. The proposed system maintains defence against typical vulnerabilities which includes both shoulder surfing weaknesses and brute-force attacks and unauthorized access protocols. Through the merge of image-based passwords with AES cryptographic algorithms and steganographic security protocols users obtain a completely secure and resistant authentication solution.

The system provides users with both simple and protected authentication methods which better protects them from traditional password vulnerabilities. The credential encryption method implemented through multiple cloud storage tools ensures data security by spreading encrypted passwords across different storage locations to protect against breaches. Encryption efficiency needs improvement along with user-experience optimization and consistent integration of platforms constitutes ongoing challenges in the system.

The system will receive future updates which aim to optimize encryption and streamline retrieval between different cloud systems and improve the login interface for enhanced usability. New developments in AI-based anomaly detection methodology combined with adaptive security protocols increase system resilience to forthcoming security threats. This secure graphical password-based application will set the standard for modern authentication systems once it evolves further because it delivers a secure yet user-friendly security solution.

VII. FUTURE ENHANCEMENT

The secure graphical password application functions better when enhanced through AI authentication analysis and optimized multi-cloud storage mechanisms which improve both security features alongside user experience. Using machine learning for real-time anomaly detection permits security systems to monitor dangerous login attempts which protects against new security threats.

The development of user verification methods should include both biometric authentication and image-based password systems because cybersecurity threats keep becoming more advanced. The development of adaptive encryption systems will improve data protection capabilities at the expense of system operational performance retention. Users can become more perceptive about security threats when the system integrates an easy-to-use dashboard which offers real-time security notifications along with login monitoring functions.

The future development of cloud storage systems requires optimization to enhance its operational speed with retention of security protocols. A new feature of multi-language support would enhance accessibility for users who speak different languages. AI authentication aids use guided security steps for users to select passwords and access their accounts securely by offering best practices.

This system will become a leader in user-centered authentication technology as encryption improvements match multi-cloud storage optimization with AI-based authentication methods.

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FORM 2 THE PATENT ACT 1970 (39 OF 1970) AND The patent rules, 2003 <u>COMPLETE SPECIFICATION</u> (See section 10: rule 13)																																			
<u>TITLE OF INVENTION</u> PFDNet: A Deep Learning Approach for Robust Shared Photo Authentication and Tamper Recovery																																			
<u>APPLICANT (S)</u> <table border="1"> <thead> <tr> <th>Name</th> <th>Nationality</th> <th>Address</th> </tr> </thead> <tbody> <tr> <td>Mr. S. Dinakar Jose</td> <td>Indian</td> <td>Assistant Professor Department of Computer Science and Engineering Anand Institute of Higher Technology Kazhipattur, Tamil Nadu, India</td> </tr> <tr> <td>Mrs. M. Maheswari</td> <td>Indian</td> <td>Head of the Department Department of Computer Science and Engineering Anand Institute of Higher Technology Kazhipattur, Tamil Nadu, India</td> </tr> <tr> <td>Mr. Giri Raj R</td> <td>Indian</td> <td>Research Scholar Department of Computer Science and Engineering Anand Institute of Higher Technology Kazhipattur, Tamil Nadu, India</td> </tr> <tr> <td>Mr. Arindran M</td> <td>Indian</td> <td>Research Scholar Department of Computer Science and Engineering Anand Institute of Higher Technology Kazhipattur, Tamil Nadu, India</td> </tr> <tr> <td>Mr. David Charles A</td> <td>Indian</td> <td>Research Scholar Department of Computer Science and Engineering Anand Institute of Higher Technology Kazhipattur, Tamil Nadu, India</td> </tr> <tr> <td>Mr. Hariharan V</td> <td>Indian</td> <td>Research Scholar Department of Computer Science and Engineering Anand Institute of Higher Technology Kazhipattur, Tamil Nadu, India</td> </tr> <tr> <td>Mr. Ajay Ganesh V</td> <td>Indian</td> <td>Research Scholar Department of Computer Science and Engineering Anand Institute of Higher Technology Kazhipattur, Tamil Nadu, India</td> </tr> <tr> <td>Mr. Chandru B</td> <td>Indian</td> <td>Research Scholar Department of Computer Science and Engineering Anand Institute of Higher Technology Kazhipattur, Tamil Nadu, India</td> </tr> <tr> <td>Mr. Gokul P</td> <td>Indian</td> <td>Research Scholar Department of Computer Science and Engineering Anand Institute of Higher Technology Kazhipattur, Tamil Nadu, India</td> </tr> <tr> <td>Mr. Jelan Albert J</td> <td>Indian</td> <td>Research Scholar Department of Computer Science and Engineering Anand Institute of Higher Technology Kazhipattur, Tamil Nadu, India</td> </tr> </tbody> </table>			Name	Nationality	Address	Mr. S. Dinakar Jose	Indian	Assistant Professor Department of Computer Science and Engineering Anand Institute of Higher Technology Kazhipattur, Tamil Nadu, India	Mrs. M. Maheswari	Indian	Head of the Department Department of Computer Science and Engineering Anand Institute of Higher Technology Kazhipattur, Tamil Nadu, India	Mr. Giri Raj R	Indian	Research Scholar Department of Computer Science and Engineering Anand Institute of Higher Technology Kazhipattur, Tamil Nadu, India	Mr. Arindran M	Indian	Research Scholar Department of Computer Science and Engineering Anand Institute of Higher Technology Kazhipattur, Tamil Nadu, India	Mr. David Charles A	Indian	Research Scholar Department of Computer Science and Engineering Anand Institute of Higher Technology Kazhipattur, Tamil Nadu, India	Mr. Hariharan V	Indian	Research Scholar Department of Computer Science and Engineering Anand Institute of Higher Technology Kazhipattur, Tamil Nadu, India	Mr. Ajay Ganesh V	Indian	Research Scholar Department of Computer Science and Engineering Anand Institute of Higher Technology Kazhipattur, Tamil Nadu, India	Mr. Chandru B	Indian	Research Scholar Department of Computer Science and Engineering Anand Institute of Higher Technology Kazhipattur, Tamil Nadu, India	Mr. Gokul P	Indian	Research Scholar Department of Computer Science and Engineering Anand Institute of Higher Technology Kazhipattur, Tamil Nadu, India	Mr. Jelan Albert J	Indian	Research Scholar Department of Computer Science and Engineering Anand Institute of Higher Technology Kazhipattur, Tamil Nadu, India
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PREAMBLE TO THE DESCRIPTION

[0001] The following specification particularly describes the invention and the manner in which it is to be performed.

DESCRIPTION

Technical field of invention:

[0002] This invention relates to the field of digital image security and authentication using deep learning techniques.

It focuses on detecting, localizing, and recovering tampered images shared across social networking platforms.

The system integrates image forensics, computer vision, and cybersecurity to ensure media integrity and prevent unauthorized modifications.

Summary of the invention:

[0003] The invention introduces PFDNet, a deep learning-based system designed for photo forgery detection and lossless recovery. It integrates modules like the Cyber Vaccinator (to make images tamper-resistant), Forgery Detector (to detect and localize image tampering), and Self-Recovery (to restore tampered regions to their original state). The system ensures high accuracy in tamper detection, seamless restoration of digital images, quality validation using Run-Length Encoding (RLE) and PSNR, and provides real-time notifications to users regarding image integrity.

Brief description of drawing:

[0004] The above and other aspects The drawing illustrates the overall system architecture of PFDNet for securing digital images on social networking platforms. It shows the workflow starting from user registration, photo upload, application of the Cyber Vaccine, tamper detection using Forgery Detector, tamper localization and mask prediction, followed by automatic recovery using an Invertible Neural Network (INN).

[0005] FIG 1 Additional components like Quality Assurance (using RLE and PSNR) and real-time notifications to users are also depicted to ensure end-to-end image authenticity and security.

Detailed description of the invention:

[0006] Forgery Detection and Localization: The system uses deep learning models like PFDNet and Invertible Neural Networks (INN) to detect image tampering, including attacks such as copy-move, splicing, and inpainting. It identifies manipulated regions with high precision and localizes them for further processing.

[0007] Self-Recovery and Quality Assurance: After detecting tampering, the system automatically restores the tampered areas using self-recovery techniques based on INN and Discrete Cosine Transform (DCT). Recovery quality is validated using Run-Length Encoding (RLE) and Peak Signal-to-Noise Ratio (PSNR) to ensure that the restored image matches the original in fidelity and integrity.

I/We claim:

1. The system claims, I wherein that claim fast recognition of Handwritten words from Image.

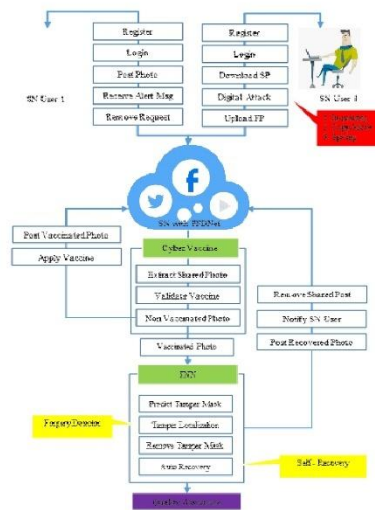


Figure 1

ABSTRACT

The invention introduces PFDNet, a deep learning-based framework for detecting tampering and recovering digital images with high fidelity. It protects images by embedding imperceptible perturbations using a Cyber Vaccinator, making them resistant to unauthorized modifications. When an image is attacked, the Forgery Detector module uses Invertible Neural Networks (INN) to predict tampered regions and localize alterations.

The system enables lossless self-recovery by reconstructing the original image using hidden perturbations and advanced restoration techniques like DCT, Run-Length Encoding (RLE) and PSNR calculations are employed to verify the quality and integrity of recovered images. The platform ensures real-time detection, localization, and restoration of tampered images shared across social networking platforms. It provides real-time alerts to users about tampering events and recovery status, enhancing transparency and trust. The solution addresses critical privacy, security, and authenticity issues associated with digital image sharing.

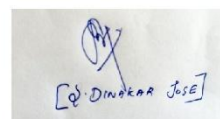
It achieves high accuracy in tamper detection even in compressed or low-resolution images. Overall, PFDNet offers a robust, scalable, and platform-independent solution for maintaining digital media integrity.

[To be published with figure 1]

THE PATENT ACT, 1970
(39 of 1970)
&
THE PATENTS RULES, 2003
REQUEST FOR PUBLICATION
[See section 11A (2) rule 24A]

I/We Mr.S.Dinakar Jose, Mrs.M.Maheswari, Mr Giri Raj R, Mr Arindran M, Mr David Charles A, Mr Haiharan V, Mr Ajay Ganesh V, Mr Chandru B, Mr Gokul P, Mr Jelan Albert J hereby request for early publication of my/our [Patent Application No.] TEMP/E-//2023CHE.

Dated this 29th day of April 2025



[S. Dinakar Jose]

Mr. S. Dinakar Jose,
Mrs. M. Maheswari,
Mr. Giri Raj R,
Mr. Arindran M,
Mr. David Charles A,
Mr. Hariharan V,
Mr. Ajay Ganesh V,
Mr. Chandru B,
Mr. Gokul P,
Mr. Jelan Albert J

To,
The Controller of Patents,
The Patent Office,
At Mumbai

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Docket Number:43545

Date/Time : 30/04/2025
Agent Number:

To,
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Sr No.	CBR No.	Reference Number /Application Type	Application Number	Title/Remarks	Amount Paid
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3		E-3/8601/2025-CHE	202541041649	Form3	0
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APPENDIX - IV
INTERNSHIP CERTIFICATES



TO
Mr. Gokul P CSE DEPARTMENT,
ANAND INSTITUTE OF HIGHER TECHNOLOGY
Date: 07 / 05 / 2025

This is to formally certify that Mr. Gokul P, a student of the Department of Computer Science and Engineering, Anand Institute of Higher Technology, has successfully completed his 3 month internship at Ajashia Technologies as part of our technical training and skill development program. During this period, he was actively engaged in Android App Development under the supervision of our experienced team. Throughout his internship, Mr. Gokul demonstrated strong commitment and enthusiasm in learning and applying his skills to real-world projects.

His role and responsibilities included:

- Understanding and applying industry practices in Android development using Java/Kotlin
- Contributing to the design, development, and testing of Android applications
- Learning about backend integration, Firebase, and modern app architecture
- Developing problem-solving and analytical skills relevant to mobile development

We appreciate his dedication and performance during the internship period. We believe that the skills and knowledge he has gained at Ajashia Technologies will greatly support his future career in the field of mobile application development.

We wish him all the best in his future endeavors.

Regards.

AJAY KUMAR .D
FOUNDER

Contact Us

8838614068

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TO
Mr. Jelen Albert J CSE DEPARTMENT,
ANAND INSTITUTE OF HIGHER TECHNOLOGY
Date: 07 / 05 / 2025

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